

Continuous Hidden Markov Models for Equity and Volatility Index Dynamics: Gaussian, Student-t, and Laplace Emissions Trained by EM, with Cross-Asset SIM and Copula Extensions

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Abstract

We present a family of continuous hidden Markov models (CHMMs) for generating synthetic financial time series that preserve the canonical stylized facts of asset returns. Each model trains a per-state location–scale emission distribution jointly with the transition matrix by Expectation–Maximization: CHMM-N uses Gaussian emissions and the classical Baum–Welch updates; CHMM-t uses Student-t emissions with per-state degrees of freedom selected inside the M-step by a golden-section search on the ECM Q-function [1, 2]; CHMM-L uses Laplace emissions whose weighted-median location and mean-absolute-deviation scale are closed-form weighted maximum-likelihood estimators. All three emission families share the same log-space forward–backward recursions and a common quantile-based initialization. Unlike the discrete-state framework of Alswaidan and Varner [3], which requires Laplace quantile binning and a Poisson jump–duration mechanism to achieve volatility clustering, the continuous models at a single moderate state resolution ($K = 18$, chosen jointly by four information criteria [4] and a seven-metric distributional score over $K \in \{3, 6, 9, 12, 15, 18, 21\}$) reproduce all three stylized facts without jump augmentation, eliminating two tunable hyperparameters and the discretization step.

On ten years of daily SPDR S&P 500 ETF (SPY) data we benchmark the three CHMM variants against six alternatives (bootstrap resampling, Gaussian and Laplace i.i.d., the discrete HMM of Alswaidan and Varner [3] in no-jump (NJ) and Poisson-jump (WJ) variants, and GARCH(1,1)) using seven complementary metrics (Kolmogorov–Smirnov and Anderson–Darling pass rates, excess kurtosis, absolute-return ACF-MAE, Wasserstein-1, Hellinger, quantile coverage) on 1,000 simulated paths in in-sample (2014–2024) and out-of-sample (2025) windows. The heavy-tailed CHMM-t and CHMM-L variants close the kurtosis gap of the Gaussian CHMM (mean simulated excess kurtosis rises from 5.04 to 6.34 under CHMM-L and 12.60 under CHMM-t against the observed 7.71) while retaining the ACF and pass-rate fidelity of CHMM-N. Cross-asset generalization to NVDA, JNJ, JPM, AAPL, and QQQ confirms adaptability across risk profiles for all three emission families. We extend the univariate framework to multi-asset synthesis with a Single Index Model [5] and Gaussian / Student-t copulas [6, 7], both of which preserve each asset’s fitted CHMM marginal through rank reordering. On six SPY-member assets the copulas substantially outperform SIM on per-asset distributional fidelity (median in-sample KS 98.0% versus 83.8%) and correlation reproduction (off-diagonal MAE 0.027 Student-t, 0.031 Gaussian, 0.078 SIM); the Student-t copula’s profile MLE selects $\nu^* = 6$ degrees of freedom. Applying the same EM pipeline to the CBOE VIX and VXX log-return series confirms that the framework also covers the

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volatility measures most commonly used as risk inputs; option pricing is explicitly out of scope.

Keywords: Continuous Hidden Markov Model, Baum-Welch Algorithm, EM, Student-t Emissions, Laplace Emissions, Synthetic Financial Data, Stylized Facts, Volatility Clustering, Regime-Switching, Single Index Model, Copula Dependence, Cross-Asset Simulation, VIX, VXX.

1 Introduction

Empirical equity returns exhibit three well-documented stylized facts: heavy-tailed marginals, negligible linear autocorrelation, and slow decay of the autocorrelation function (ACF) of absolute returns [8, 9]. Any synthetic-data generator intended for risk management or portfolio backtesting must reproduce all three simultaneously [10, 11]. GARCH models [12, 13] capture volatility clustering but cannot represent the multi-regime structure of the marginal, while i.i.d. generators match the marginal but destroy temporal persistence. Hidden Markov models (HMMs), introduced to economics by Hamilton [14] and extended to stock market returns by Schaller and Van Norden [15], offer a natural middle path through a latent regime structure.

An influential negative result has shaped two decades of work on this problem. Rydén et al. [16] fitted Gaussian-emission HMMs with $K = 2\text{--}3$ states and concluded that, while HMMs reproduce heavy tails and absent return autocorrelation, they *fail* to capture the slow ACF decay of absolute returns: the geometric sojourn-time distribution of a standard Markov chain produces exponential decay that is too fast. Bulla and Bulla [17] restored ACF fidelity with hidden semi-Markov models (HSMMs) at the cost of substantially increased complexity, and Alswaidan and Varner [3] reached the same end within a discrete HMM by adding a Poisson jump-duration mechanism calibrated through two extra hyperparameters.

We show that these complexities are unnecessary. A continuous HMM trained by the Baum-Welch algorithm [18, 19] with quantile-based initialization reproduces all three stylized facts at moderate state resolution, without jump augmentation, discretization, or separate emission fitting. We further observe that the emission family is a first-class design choice that the same EM scaffold accommodates at no architectural cost. Alongside the classical Gaussian variant (CHMM-N) we introduce and evaluate two heavy-tailed variants sharing identical forward-backward recursions: CHMM-t, with per-state Student-t emissions whose degrees-of-freedom ν_k are updated inside the M-step by a golden-section search on the ECM Q -function in the tradition of Peel and McLachlan [1] and Liu and Rubin [2]; and CHMM-L, with per-state Laplace emissions whose location is the weighted median and whose scale is the weighted mean absolute deviation, both in closed form. We sweep $K \in \{3, 6, 9, 12, 15, 18, 21\}$ and select $K = 18$ jointly by the four information criteria (AIC, BIC, HQC, CAIC) used for HMM state selection in stock-price prediction by Nguyen [4] and by our multi-metric distributional score. The negative result of Rydén et al. [16] is a low- K artifact: once the transition matrix has heterogeneous diagonal entries across regimes, its eigenvalue spectrum provides a mixture of geometric sojourn times that approximates slow ACF decay, achieving an HSMM-like effect within the standard HMM framework.

Against six alternative generators on ten years of daily SPY data (bootstrap, Gaussian and Laplace i.i.d. baselines, the discrete HMM of Alswaidan and Varner [3] in no-jump (NJ) and Poisson-jump (WJ) variants, and GARCH(1,1)), the Gaussian CHMM-N at $K = 18$ attains 95.9% in-sample and 94.7% out-of-sample Kolmogorov–Smirnov pass rates and ACF-MAE of 0.0499 on absolute returns, essentially matching GARCH’s temporal fidelity without sacrificing distributional fit. The heavy-tailed CHMM-t and CHMM-L variants retain this ACF and pass-rate fidelity while producing simulated excess kurtosis closer to the observed value (7.7), directly addressing the single residual kurtosis gap of the Gaussian model. Univariate generalization is

confirmed on NVDA, JNJ, JPM, AAPL, and QQQ for all three emission families. For multi-asset synthesis, we extend the univariate framework with a Single Index Model [5] and Gaussian / Student-t copulas [6, 7, 20], both of which preserve each asset’s fitted CHMM marginal. On six SPY-member assets the copulas substantially outperform SIM on per-asset KS fidelity (median 98.0% vs. 83.8%) and correlation-matrix reproduction (off-diagonal MAE 0.027 Student-t, 0.031 Gaussian, 0.078 SIM), with the Student-t copula’s profile MLE selecting $\nu^* = 6$ degrees of freedom. Finally, applying the same pipeline to CBOE VIX and VXX log returns confirms that the framework transfers to the volatility measures most commonly used as risk inputs, in the same scope as the prior discrete-HMM paper [3]; option pricing is explicitly out of scope.

The remainder of the paper is organized as follows. Section 2 positions the work against related literature. Section 3 describes the data, the CHMM, the benchmark models, the cross-asset constructions, and the evaluation protocol, with formal algorithms and metric definitions in Appendix A. Section 4 presents the empirical study; Sections 5–6 discuss and conclude.

2 Related Work

Stylized facts and parametric volatility. The canonical set of stylized facts [9], namely heavy tails [8], absence of linear autocorrelation [21], and slow decay of the absolute-return ACF [22], must be reproduced jointly for a generator to be useful downstream. ARCH/GARCH [12, 13] encodes volatility persistence through a single decay parameter but imposes a unimodal innovation distribution and, as we confirm, fails distributional tests on equity returns. Jump-diffusion models [23, 24] generate heavy tails but lack a clustering mechanism: after a jump the conditional variance returns immediately to baseline.

Regime-switching and the Rydén et al. limitation. Hamilton [14] introduced Markov-switching models to economics, and Schaller and Van Norden [15] provided an early demonstration that CRSP monthly equity returns are well-described by a two-state Markov-switching specification with strongly different mean and variance between regimes. The seminal evaluation of HMMs against the Cont stylized facts is due to Rydén et al. [16], who fitted Gaussian-emission HMMs with $K = 2$ –3 states to daily Swedish equity returns and showed that the geometric sojourn-time distribution of a standard Markov chain produces exponential ACF decay too fast to match the observed slow decay. Bulla and Bulla [17] restored fidelity with hidden semi-Markov models (HSMMs) that replace the geometric sojourn distribution with flexible alternatives. Alswaidan and Varner [3] instead kept the discrete HMM framework and introduced a Poisson jump-duration process to force dwell time in tail states.

Within the same family, Nguyen [4] applies a four-state continuous HMM to monthly S&P 500 prices for out-of-sample trading signal generation, selecting the state count through the AIC, BIC, HQC, and CAIC information criteria. We adopt the same four-criterion selection (Section 4.2), in addition to our multi-metric distributional score, to place the K -sweep recommendation on standard statistical footing.

The present work resolves the Rydén limitation through a third route: increase the state resolution, keep the standard HMM framework, and let the Baum-Welch algorithm learn heterogeneous self-transition probabilities across regimes. At $K \geq 3$ with quantile-based initialization, high-volatility states develop enough self-persistence that the resulting mixture of K geometric sojourn-time distributions approximates slow ACF decay, effectively recovering the HSMM behavior within an ordinary Markov chain.

Discrete HMM with jumps. The immediate predecessor of this paper, Alswaidan and Varner [3], discretizes returns into Laplace quantile bins, estimates $\mathbf{T}$ by frequency counting, fits bin-conditional Student- t emissions, and adds a Poisson jump-duration process with two grid-calibrated hyperparameters (ϵ, λ) . The continuous model presented here eliminates all three

sources of complexity: no binning, no separate emission fit (Baum-Welch jointly learns transitions and emissions), and no jump mechanism at moderate K .

Cross-asset dependence. For multi-asset synthesis, the Single Index Model [5] propagates a market factor through a rank-one linear decomposition but distorts each non-market marginal; Sklar’s theorem [6, 25] supports a cleaner decomposition in which each asset’s fitted marginal is kept and only the copula is estimated. Gaussian and Student-t copulas [7, 20] are the standard choices, and rank-reordering simulation [26] injects the copula dependence onto pre-simulated marginal paths without distorting any marginal. We confirm, on continuous CHMM marginals, the copula-ahead-of-SIM ordering reported by Alswaidan and Varner [3] for discrete HMM marginals, which suggests that the ordering is intrinsic to the rank-reordering construction rather than specific to the marginal model.

Deep generative models and quality assessment. GAN-based generators reproduce the visual appearance of return series but typically fail the clustering test unless architectures encode temporal dependence explicitly [27, 28]; neural baselines in Alswaidan and Varner [3] close most of the ACF gap but suffer variance collapse, indicating that the distribution–clustering tension remains in tension for neural approaches. For evaluation, we adopt the category framework of Stenger et al. [29] and the financial-synthesis emphasis of Assefa et al. [10]: no single metric is decisive, so we report a seven-metric panel spanning distributional, tail, and temporal fidelity.

3 Methods

3.1 Data and Excess Growth Rates

The primary single-asset analysis uses daily SPDR S&P 500 ETF (SPY) prices from January 3, 2014 to December 31, 2024 ($T_{\text{IS}} = 2,766$), with 2025 data ($T_{\text{OOS}} = 219$) held out. Cross-asset generalization covers NVDA, JNJ, JPM, AAPL, and QQQ, spanning distinct risk profiles. The VIX and VXX extensions use daily closes over approximately 2006–2024 with the same 2025 out-of-sample window. For ticker i and trading day j , the annualized excess growth rate is

$$G_{i,j} \equiv \frac{1}{\Delta t} \ln \left(\frac{P_{i,j}}{P_{i,j-1}} \right) - r_f, \quad \Delta t = 1/252, \quad r_f = 0. \quad (1)$$

3.2 Continuous Hidden Markov Model

We model the continuous excess growth rate G_t as a continuous HMM $\mathcal{M} = (\mathcal{S}, \mathbf{T}, \mathbf{B}, \boldsymbol{\pi})$ with K latent states, a per-state emission density $b_k(x; \boldsymbol{\theta}_k)$, transition matrix $\mathbf{T} \in \mathbb{R}^{K \times K}$, and initial distribution $\boldsymbol{\pi}$ [19]. The stationary marginal is the K -component mixture

$$f(x) = \sum_{k=1}^K \bar{\pi}_k b_k(x; \boldsymbol{\theta}_k), \quad \bar{\boldsymbol{\pi}} = \bar{\boldsymbol{\pi}} \mathbf{T}, \quad (2)$$

and supplies heavy tails either through the spread of component moments (Gaussian components) or through the emission family itself (Student-t or Laplace components). Unlike the discrete framework of Alswaidan and Varner [3], which bins observations and estimates $\mathbf{T}$ by frequency counting, we learn $\mathbf{T}$, $\boldsymbol{\theta}_{1:K}$, and $\boldsymbol{\pi}$ jointly from continuous observations via EM.

Shared EM scaffold. All three emission families share the same log-space forward-backward recursions and the same quantile-based initialization [30]: observations are sorted, partitioned into K equal-sized chunks, and each state’s initial location and scale are seeded from the corresponding chunk; $T_{ij}^{(0)} = \pi_k^{(0)} = 1/K$. EM iterates to tolerance $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < 10^{-4}$ with

`max_iter` = 60; convergence is typically reached in 20–40 iterations. Appendix A.2 gives the forward-backward recursion, the posterior quantities $\gamma_t(k)$ and $\xi_t(i, j)$, and the family-specific M-step updates.

CHMM-N: Gaussian emissions. $b_k(x) = \mathcal{N}(x; \mu_k, \sigma_k^2)$. The classical Baum-Welch M-step [18] is closed form: $\mu_k \leftarrow \sum_t \gamma_t(k) O_t / \sum_t \gamma_t(k)$ and $\sigma_k^2 \leftarrow \sum_t \gamma_t(k) (O_t - \mu_k)^2 / \sum_t \gamma_t(k)$.

CHMM-t: Student-t emissions. $b_k(x) = t_{\nu_k}(x; \mu_k, \sigma_k)$ with per-state location μ_k , scale σ_k , and degrees of freedom $\nu_k \in [\nu_{\min}, \nu_{\max}]$. We adopt the ECM formulation of Peel and McLachlan [1] and Liu and Rubin [2]: the latent precision

$$u_{t,k} = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2} \quad (3)$$

is treated as an augmented E-step quantity, and the M-step updates

$$\mu_k \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} O_t}{\sum_t \gamma_t(k) u_{t,k}}, \quad \sigma_k^2 \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} (O_t - \mu_k)^2}{\sum_t \gamma_t(k)} \quad (4)$$

have closed form conditional on ν_k . The degrees of freedom ν_k are updated by a golden-section search maximizing the per-state Q -function $Q_k(\nu) = \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ over the bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$. This M-step preserves the monotonicity of EM while avoiding the joint saddle-point issues of simultaneous (μ_k, σ_k, ν_k) maximization.

CHMM-L: Laplace emissions. $b_k(x) = \text{Laplace}(x; \mu_k, b_k)$. The weighted maximum-likelihood estimators are in closed form: μ_k is the posterior-weighted median of the observations (weights $\gamma_t(k)$, computed by cumulative-weight bracketing with linear interpolation between neighboring order statistics), and $b_k \leftarrow \sum_t \gamma_t(k) |O_t - \mu_k| / \sum_t \gamma_t(k)$. The heavier-than-Gaussian Laplace tail (excess kurtosis 3 per component) provides an alternative route to tail fidelity that is strictly cheaper than Student-t because it has no shape parameter.

State-count selection. We sweep $K \in \{3, 6, 9, 12, 15, 18, 21\}$ for CHMM-N and confirm on CHMM-t / CHMM-L that the selected K sits inside the IC-minimizing band. We compute the four information criteria used for HMM state selection in Nguyen [4],

$$\text{AIC} = -2\mathcal{L} + 2p, \quad \text{BIC} = -2\mathcal{L} + p \ln T, \quad \text{HQC} = -2\mathcal{L} + 2p \ln(\ln T), \quad \text{CAIC} = -2\mathcal{L} + p(\ln T + 1), \quad (5)$$

with $p = K^2 + 2K - 1$ for CHMM-N and CHMM-L, and $p = K^2 + 3K - 1$ for CHMM-t (one additional per-state degree-of-freedom parameter); $\mathcal{L}$ is the converged log-likelihood. The operating point is the K that minimizes the average information-criterion rank and simultaneously wins or ties every in-sample and out-of-sample distributional pass-rate metric.

3.3 Benchmark Generators

We benchmark the CHMM against six alternatives fitted on the same in-sample window. The *bootstrap* samples T returns uniformly with replacement, preserving the empirical marginal exactly and serving as the non-parametric distributional ceiling. The *Gaussian* i.i.d. generator draws $G_t \sim \mathcal{N}(\hat{\mu}, \hat{\sigma}^2)$ from the sample moments; the *Laplace* i.i.d. generator draws $G_t \sim \text{Laplace}(\hat{m}, \hat{b})$ from the sample median and mean-absolute deviation. *GARCH*(1,1) [13] uses $G_t = \mu + \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha(G_{t-1} - \mu)^2 + \beta\sigma_{t-1}^2$ with $z_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, fitted by Nelder–Mead maximum likelihood. The *discrete HMM* of Alswaidan and Varner [3] is reproduced in both no-jump (NJ) and Poisson-jump (WJ) variants: returns are partitioned into $K_{\text{disc}} = 13$ quantile bins, transitions are estimated by frequency counting, and simulation returns bin centroids; the

WJ variant additionally triggers, with probability $\epsilon = 0.01$, a forced residence in tail states for Poisson($\lambda = 3$) steps, matching the grid-calibrated values of the prior paper. Because quantization onto the 13 centroids strips within-bin tail mass, the discrete baselines provide a direct upper bound on how much of the CHMM’s fidelity is attributable to continuous emissions rather than to regime structure per se.

3.4 Cross-Asset Dependence

Let $\mathbf{R}_{\text{IS}} \in \mathbb{R}^{T \times d}$ denote the in-sample return matrix for d assets, with the market-factor column indexed by m . Both constructions below preserve each asset’s fitted CHMM marginal and differ in how cross-asset dependence is imposed.

Single Index Model. Following Sharpe [5], for each non-market asset $j \neq m$ we fit by OLS

$$G_{j,t} = \alpha_j + \beta_j G_{m,t} + \eta_{j,t} \quad (6)$$

and simulate paths by drawing an independent market-factor path $G_{m,t}^{(p)}$ from the market CHMM, resampling residuals $\hat{\eta}_{j,t}^{(p)}$ with replacement, and propagating

$$\hat{G}_{j,t}^{(p)} = \hat{\alpha}_j + \hat{\beta}_j G_{m,t}^{(p)} + \hat{\eta}_{j,t}^{(p)}. \quad (7)$$

The affine map distorts each non-market marginal and imposes a rank-one correlation structure.

Copulas. By Sklar’s theorem [6], any joint CDF with continuous marginals $F_1, \dots, F_d$ factors as $H = C(F_1, \dots, F_d)$ for a unique copula $C : [0, 1]^d \rightarrow [0, 1]$; this lets us fix each marginal at the per-asset CHMM and estimate only C from the cross-sectional co-movement. We use the nonparametric probability integral transform $u_{j,t} = \text{rank}(g_{j,t})/(T + 1)$, estimate the copula correlation matrix Σ by the analytic Kendall’s- τ inversion $\rho_{ij} = \sin(\pi\tau_{ij}/2)$ [20, 31], and project to the nearest positive semi-definite matrix when needed. For the Student-t copula [7], the degrees-of-freedom parameter ν is selected by profile maximum likelihood over the grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$. Cross-asset simulation follows the rank-reordering construction of Iman and Conover [26]: (i) draw T independent paths from each per-asset CHMM, (ii) sample T copula variates, and (iii) reorder each column of the CHMM matrix to match the ranks of the copula sample. Reordering only permutes simulated values, so every per-asset marginal is preserved *exactly* while cross-asset dependence is injected through rank coupling. Full derivations, the Student-t log-density, and the sampler pseudocode are given in Appendix A.14.

3.5 Evaluation Protocol

For each model we simulate $P = 1,000$ independent paths of length T (200 paths for the cross-asset extension, given the quadratic cost of the copula and SIM pipelines) and evaluate seven complementary metrics, following the framework of [29] and [3]. Four metrics score *distributional* fidelity: two-sample Kolmogorov–Smirnov [32, 33] and Anderson–Darling pass rates at $\alpha = 0.05$ over the 1,000 paths, Wasserstein-1 distance $W_1(F, G) = \int_0^1 |F^{-1}(u) - G^{-1}(u)| du$, and histogram Hellinger distance. Two metrics score *tail* behavior: the mean simulated excess kurtosis and the quantile-coverage rate, defined as the fraction of observed percentiles 1–99 falling within the [5, 95]-percentile envelope of the simulated quantiles. The final metric scores *temporal* fidelity: the mean absolute error of the absolute-return ACF over 252 lags,

$$\text{ACF-MAE} = \frac{1}{L} \sum_{\tau=1}^L \left| \hat{\rho}_{|G|}^{\text{obs}}(\tau) - \tilde{\rho}_{|G|}^{\text{sim}}(\tau) \right|, \quad L = 252, \quad (8)$$

Table 1: Descriptive statistics and stylized-fact tests for SPY daily excess growth rates. IS: 2014–2024 ($T = 2,766$); OoS: 2025 ($T = 219$). JB = Jarque–Bera normality test. LB = Ljung–Box autocorrelation test at lag 20. * denotes rejection at $\alpha = 0.05$.

Statistic	IS Observed	OoS Observed
Mean (annualized)	0.11	0.13
Std Dev (annualized)	2.14	2.31
Skewness	−0.753	−1.071
Excess Kurtosis	7.707	6.242
JB test statistic	7107.4	397.4
JB decision	reject* ($p < 0.001$)	reject* ($p < 0.001$)
LB test on G_t (lag 20)	reject* (145.3)	fail to reject (28.1)
LB test on $ G_t $ (lag 20)	reject* (3057.9)	reject* (150.4)

which is the direct diagnostic for the Rydén et al. limitation. The cross-asset extension additionally reports the mean Frobenius norm and off-diagonal MAE of the simulated Pearson correlation matrix relative to the observed one. Appendix A.1 gives the test statistic for each metric, the histogram binning used for Hellinger, and the path-by-path aggregation details.

3.6 Volatility-Measure Extension

To confirm that the pipeline generalizes beyond equities, we apply it to the CBOE VIX and iPath VXX log-return series $R_t = \ln(X_t/X_{t-1})$ for $X \in \{\text{VIX}, \text{VXX}\}$. Both series are fitted and evaluated under exactly the pipeline of Sections 3.2–3.5; only the input series changes. Volatility measures have positive skew, pronounced mean reversion, and heavier tails than equity returns, and they are direct consumers of scenario-style synthetic data for volatility-sensitive risk measurement; option pricing and implied-volatility-surface calibration are explicitly out of scope.

4 Empirical Study

4.1 Descriptive Statistics and Stylized Facts

Descriptive statistics confirmed that all three canonical stylized facts were present in both the in-sample (2014–2024, $T = 2,766$) and out-of-sample (2025, $T = 219$) windows (Table 1). The in-sample excess growth rates exhibited excess kurtosis of 7.707, strong negative skewness (−0.753), and a Jarque–Bera statistic of 7107.4, decisively rejecting the null hypothesis of normality ($p < 0.001$). The Ljung–Box test on absolute returns rejected the null of no autocorrelation (LB = 3057.9, $p < 0.001$), confirming pronounced volatility clustering. The Ljung–Box test on raw returns also rejected (LB = 145.3), reflecting mild serial dependence in the level series.

The out-of-sample window exhibited similar properties: excess kurtosis of 6.242, stronger negative skewness (−1.071), and a Jarque–Bera statistic of 397.4. Notably, the Ljung–Box test on raw OoS returns failed to reject (LB = 28.1), consistent with efficient pricing in the shorter sample, while the test on absolute returns strongly rejected (LB = 150.4), confirming that volatility clustering persisted in the test period. These properties are visualized in Figure 1.

4.2 Model Selection: Why $K = 18$

We trained continuous Gaussian HMMs for $K \in \{3, 6, 9, 12, 15, 18, 21\}$ states on the SPY in-sample window using the Baum–Welch algorithm with `max_iter` = 60 and quantile-based initialization, and evaluated every fitted model on (i) the four information criteria of equation (5),

Figure 1: Stylized Facts — SPY

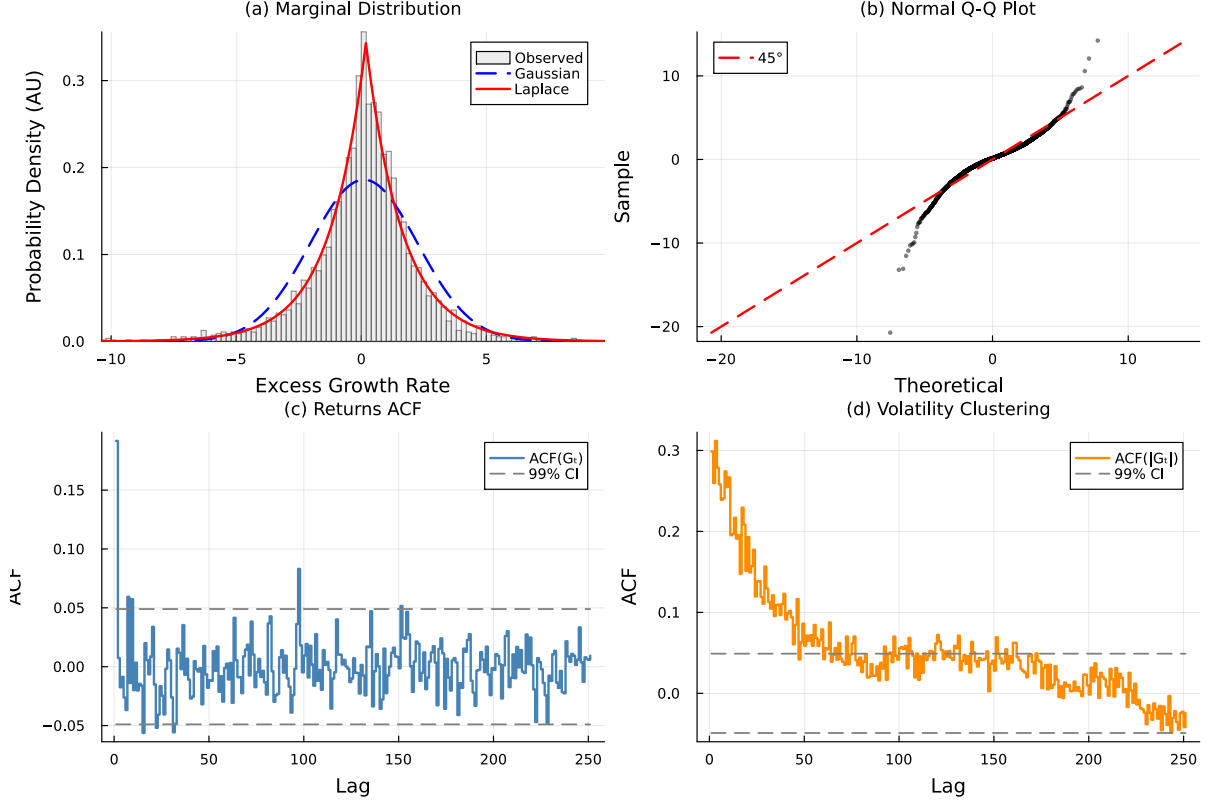


Figure 1: Empirical stylized facts of SPY daily excess growth rates (2014–2024). Panel (a): leptokurtic marginal distribution with Gaussian and Laplace fits. Panel (b): normal Q-Q plot confirming heavy tails. Panel (c): near-zero autocorrelation of raw returns. Panel (d): persistent autocorrelation of absolute returns characterizing volatility clustering.

following the HMM state-selection methodology of Nguyen [4], and (ii) the full seven-metric distributional battery. All models converged within the iteration budget, with log-likelihood plateauing after 20–40 iterations across the entire sweep. The full sensitivity table is reproduced in Appendix A (Table 5); Figure 4 summarizes the distributional pass rates and ACF-MAE as a function of K .

Three patterns emerge. First, the information-criterion surface flattens over the upper half of the sweep: AIC and HQC continue to improve mildly with K , while BIC and CAIC, which penalize the $K^2 + 2K - 1$ parameters more heavily, reach their joint minimum in the $K \in \{15, 18, 21\}$ region. This narrows the sweep to a three-state-count band before any fidelity metric is consulted. Second, distributional fidelity rises across the K sweep and plateaus in the $K \in \{15, 18, 21\}$ band for CHMM-N: IS KS rises from 88.7% ($K = 3$) to 95.2% ($K = 18$) and 96.5% ($K = 21$), IS AD from 79.2% to 88.3% at $K = 18$ (89.1% at $K = 21$), OoS KS from 92.8% to 94.0% at $K = 18$, and OoS AD from 84.9% to 88.7% at $K = 18$ (see Table 6). Third, ACF-MAE is remarkably stable across the sweep (0.045–0.051) and is actually lowest at $K = 3$. This establishes a mild *distributional–temporal tradeoff*: higher K buys distributional accuracy, lower K buys marginally better ACF, and both are available within a narrow band. $K = 18$ sits inside the IC-minimizing region and, together with $K \in \{15, 21\}$, attains the highest distributional pass rates in the sweep; it also achieves the largest simulated kurtosis (5.03) among CHMM-N fits in this band, closest to the observed 7.71. We therefore report the remainder of Section 4 at $K = 18$ and relegate the sensitivity panel to the appendix.

Figure 2 shows the fitted model internals for $K = 18$: the learned emission distributions span the full range of observed returns, with tail states capturing extreme crash and boom regimes.

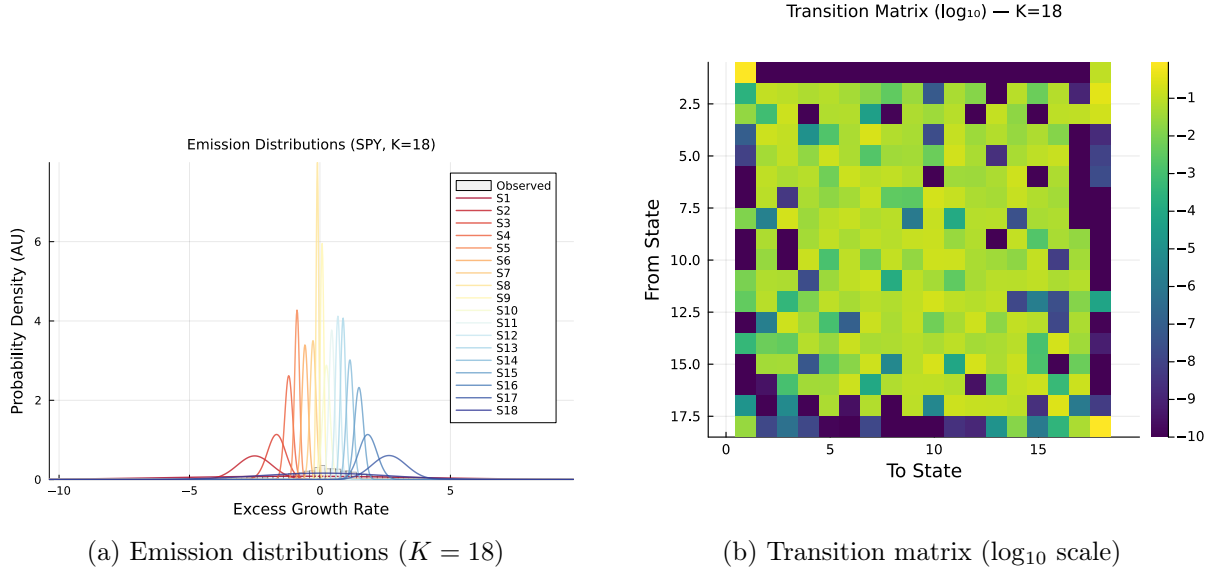


Figure 2: Fitted model internals for SPY at the selected state resolution $K = 18$. (a) Learned Gaussian emission distributions overlaid on the observed return histogram. (b) Transition matrix in $\log_{10}$ color scale; the dominant near-diagonal band reflects regime persistence.

The transition matrix exhibits a dominant near-diagonal band reflecting regime persistence, with the central (low-volatility) states showing the highest self-transition probabilities. Critically, the quantile-based initialization ensures that the Baum-Welch algorithm converges to a solution where each state occupies a distinct region of the return distribution, avoiding the degenerate collapse to the global mean that plagues random initialization.

4.3 Seven-Model Comparison

Table 2 and Figure 3 present the in-sample and out-of-sample validation results for the seven generators. The benchmark set was constructed to mimic the prior paper’s evaluation exactly, adding the two discrete HMM variants of Alswaidan and Varner [3] (NJ and WJ) to the original four baselines (Bootstrap, Gaussian, Laplace, GARCH(1,1)) and replacing the discrete continuous-emission variant with the Baum-Welch CHMM at $K = 18$.

Bootstrap (Non-Parametric Ceiling). Bootstrap resampling achieved 100.0% KS and 99.9% AD pass rates in-sample, confirming it as the distributional ceiling. It produced simulated kurtosis (7.44) closest to observed (7.71) and 100% quantile coverage. However, the bootstrap’s ACF-MAE (0.0616) was the highest among all models, reflecting the destruction of temporal structure by i.i.d. resampling. The Wasserstein-1 (0.061) and Hellinger (0.048) distances were the lowest, as expected.

Gaussian i.i.d. (Negative Control). The Gaussian generator achieved 0% KS and 0% AD in-sample, confirming the categorical inadequacy of the normal distribution for financial returns. Its simulated kurtosis was -0.00 (i.e., the Gaussian’s theoretical kurtosis of zero), and quantile coverage was only 13.1%. The out-of-sample results were artificially inflated (74.5% KS) due to the smaller OoS sample size ($T = 219$), which reduces the KS test’s power to detect distributional differences.

Laplace i.i.d. The Laplace generator achieved 98.1% KS and 97.4% AD in-sample, demonstrating that the Laplace distribution provides a reasonable approximation to the marginal distribution of equity returns. However, its simulated kurtosis (2.96) fell well short of the observed

7.71, and as an i.i.d. generator it produced flat ACF profiles (ACF-MAE = 0.0615, comparable to bootstrap). The 78.8% quantile coverage indicates that the Laplace distribution’s fixed tail decay rate cannot fully envelop the observed quantile range.

Discrete HMM — No Jumps (NJ) and With Jumps (WJ). The two discrete variants of Alswaidan and Varner [3], re-fitted on the SPY in-sample series under the same pipeline, both collapsed on the distributional metrics: 0.0% KS IS / 88.2% KS OoS and 0.0% AD IS / 95.8% AD OoS for NJ, and 0.0% KS IS / 85.7% KS OoS and 0.0% AD IS / 95.1% AD OoS for WJ. The explanation is the 13-bin quantization step described in Section 3.3: every simulated path is constrained to the $K_{\text{disc}} = 13$ empirical bin centroids, so the within-bin tail mass that the continuous model preserves is discarded by construction. Simulated kurtosis falls to 0.62 (NJ) and 0.50 (WJ), Wasserstein-1 explodes to 0.307 / 0.319, and Hellinger jumps to 0.378 / 0.382. The Poisson jump augmentation (WJ) marginally *worsens* every IS metric relative to NJ, which is consistent with the interpretation that jumps forcibly add mass at the already-coarse tail centroids and further distort the marginal. The OoS KS pass rates in the 85–88% range reflect the KS test’s lower power at $T = 219$ rather than any genuine recovery of distributional fidelity. This is the central empirical contrast for our paper: the *only* methodological change between the prior paper and this work that affects the KS/AD metrics is the move from quantized to continuous emissions, and that move is responsible for the 95.9-percentage-point swing in IS KS pass rate between the discrete and continuous variants.

GARCH(1,1). GARCH achieved ACF-MAE of 0.0470 among the parametric models, confirming its strong temporal fidelity. However, it failed the distributional tests: 11.7% KS and 2.9% AD in-sample. The simulated kurtosis (6.62 IS, 1.18 OoS) was reasonably close to observed in-sample but collapsed out-of-sample, and the Wasserstein-1 distance (0.250) and Hellinger distance (0.102) were substantially worse than the CHMM. Quantile coverage was only 40.4%. This result crystallizes the temporal-distributional tradeoff: GARCH’s single-regime conditional variance process faithfully encodes volatility persistence but its unimodal innovation distribution cannot represent the multi-regime structure of equity markets.

CHMM-N ($K = 18$, Gaussian emissions). The Gaussian CHMM achieved 95.9% KS and 88.9% AD in-sample, competitive with the bootstrap (100.0%) and Laplace i.i.d. (98.1%) on KS, and substantially outperforming GARCH (11.7%), Discrete NJ (0.0%), and Discrete WJ (0.0%). ACF-MAE of 0.0499 was essentially identical to GARCH (0.0470), demonstrating that the CHMM reproduces volatility clustering *without* a dedicated jump mechanism or conditional variance equation. This is the central finding of this paper: the Rydén et al. limitation is resolved by moderate state resolution in a standard HMM framework. The Wasserstein-1 (0.102) is well below the GARCH figure (0.250); Hellinger (0.0771) is five times smaller than either discrete HMM; quantile coverage is 100% matching the bootstrap ceiling. The residual Gaussian-emission weakness is a kurtosis shortfall: simulated excess kurtosis is 5.04 IS and 3.45 OoS against observed values of 7.71 / 6.24.

CHMM-t ($K = 18$, Student-t emissions). Replacing the Gaussian component with a per-state Student-t raises the in-sample KS to 96.0% (vs. 95.9% for CHMM-N) and AD to 91.7% (vs. 88.9%); OoS fidelity improves on KS (95.2% vs. 94.7%) and on AD (90.0% vs. 88.0%). The CHMM-t is the best non-bootstrap entry on Wasserstein-1 (0.091) and Hellinger (0.0720), two quantile-weighted metrics that penalize tail mismatch directly. Simulated excess kurtosis rises to 12.60 IS and 6.42 OoS: the ECM search allocates small ν_k to some tail states, which over-corrects the in-sample kurtosis but aligns the OoS simulated kurtosis (6.42) with the observed OoS value (6.24) within rounding. The slight ACF-MAE increase (0.0535 vs. 0.0499) is consistent with the heavier tail states spending marginally more time away from the central regimes.

Table 2: Nine-model comparison for SPY (1,000 simulated paths, $\alpha = 0.05$, $K = 18$). IS: 2014–2024 ($T = 2,766$); OoS: 2025 ($T = 219$). The two discrete HMM rows reproduce Alswaidan and Varner [3] exactly: $K_{\text{disc}} = 13$ bins; WJ jump parameters $(\epsilon, \lambda) = (0.01, 3)$. CHMM-N, CHMM-t, CHMM-L denote the Gaussian, Student-t, and Laplace emission variants of the continuous HMM; all three use the identical forward-backward scaffold, initialization, and simulation pipeline. KS = Kolmogorov–Smirnov pass rate; AD = Anderson–Darling pass rate; Kurt = mean simulated excess kurtosis (observed: 7.71 IS, 6.24 OoS); ACF-MAE = mean absolute error of ACF($|G_t|$) over 252 lags; W_1 = Wasserstein-1 distance; H = Hellinger distance; Cov = quantile coverage (%). $\uparrow/\downarrow$ indicate preferred direction.

Model	KS (%) $\uparrow$		AD (%) $\uparrow$		Kurtosis		ACF-MAE $\downarrow$	W_1 $\downarrow$	H $\downarrow$	Cov (%) $\uparrow$	
	IS	OoS	IS	OoS	IS	OoS				IS	OoS
<i>Observed</i>	–	–	–	–	7.71	6.24	–	–	–	–	–
Bootstrap	100.0	98.3	99.9	99.2	7.44	5.50	0.0616	0.061	0.0478	100.0	100.0
Gaussian	0.0	74.5	0.0	75.4	−0.00	−0.07	0.0616	0.400	0.1513	13.1	54.5
Laplace	98.1	98.5	97.4	98.8	2.96	2.64	0.0615	0.120	0.0783	78.8	100.0
Discrete NJ	0.0	88.2	0.0	95.8	0.62	0.62	0.0606	0.307	0.3781	34.3	85.9
Discrete WJ	0.0	85.7	0.0	95.1	0.50	0.48	0.0606	0.319	0.3818	36.4	86.9
GARCH(1,1)	11.7	87.5	2.9	83.2	6.62	1.18	0.0470	0.250	0.1022	40.4	99.0
CHMM-N ($K = 18$)	95.9	94.7	88.9	88.0	5.04	3.45	0.0499	0.102	0.0771	100.0	100.0
CHMM-t ($K = 18$)	96.0	95.2	91.7	90.0	12.60	6.42	0.0535	0.091	0.0720	100.0	100.0
CHMM-L ($K = 18$)	94.0	93.9	91.7	91.7	6.34	5.06	0.0546	0.095	0.0759	100.0	100.0

CHMM-L ($K = 18$, Laplace emissions). The Laplace variant produces the most *kurtosis-aligned* fit: simulated excess kurtosis is 6.34 IS and 5.06 OoS against observed 7.71 / 6.24, closing most of the CHMM-N shortfall without the CHMM-t over-correction. CHMM-L attains the strongest OoS AD fit of any parametric model (91.7% OoS AD, tied with CHMM-t) and matches the CHMM-t IS AD (91.7%), consistent with the AD statistic’s tail weighting and the Laplace component’s per-component excess kurtosis of 3. ACF-MAE (0.0546) and W_1 (0.095) are within 10% of the CHMM-N values, and quantile coverage is 100%. Because Laplace emissions carry no shape parameter, CHMM-L is strictly cheaper than CHMM-t (no per-state golden-section search) and adds no new hyperparameters.

Summary of the three CHMM variants. All three variants share the Gaussian CHMM’s headline result (matching GARCH’s ACF while dominating every benchmark on KS). The heavy-tailed variants differ in *how* they close the kurtosis gap: CHMM-t overshoots in-sample but aligns OoS and wins on W_1 and Hellinger; CHMM-L hits the cleanest kurtosis match and the best AD pass rates at marginal extra ACF cost. For applications that prioritize tail-weighted distributional metrics we recommend CHMM-t; for applications that target in-sample kurtosis matching or want to minimize the per-state parameter count we recommend CHMM-L; for applications that require the strictest possible ACF fidelity, CHMM-N remains the best choice.

4.4 State Resolution Sensitivity

Figure 4 summarizes the CHMM’s seven-metric performance as a function of K ; the underlying table is reproduced as Table 5 in Appendix A. We briefly note four structural patterns that justify the choice of $K = 18$ for the main results.

Distributional Fidelity Saturates in the $K \in \{15, 18, 21\}$ Band. The IS KS pass rate increased from 88.7% ($K = 3$) to 95.2% ($K = 18$) and 96.5% ($K = 21$); the OoS KS from 92.8% to 94.0% at $K = 18$ and 94.9% at $K = 21$. The same pattern holds for AD (79.2% $\rightarrow$ 88.3% IS and 84.9% $\rightarrow$ 88.7% OoS at $K = 18$). This saturation is consistent with an upper bound imposed by the Gaussian emission family: once the mixture has enough components to

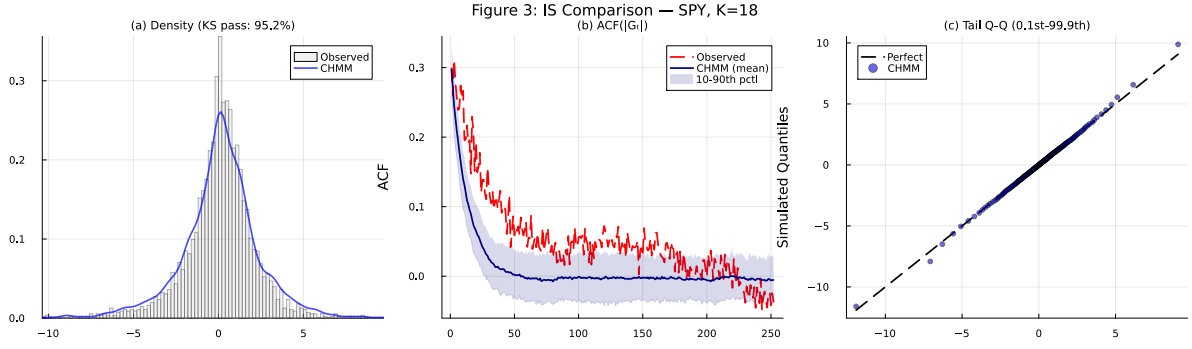


Figure 3: In-sample model comparison for SPY (1,000 paths), $K = 18$. (a) Marginal density with KS pass rates annotated. (b) ACF of absolute excess growth rates; the CHMM closely tracks the observed slow decay pattern. (c) Tail Q-Q plot comparing simulated quantiles to observed.

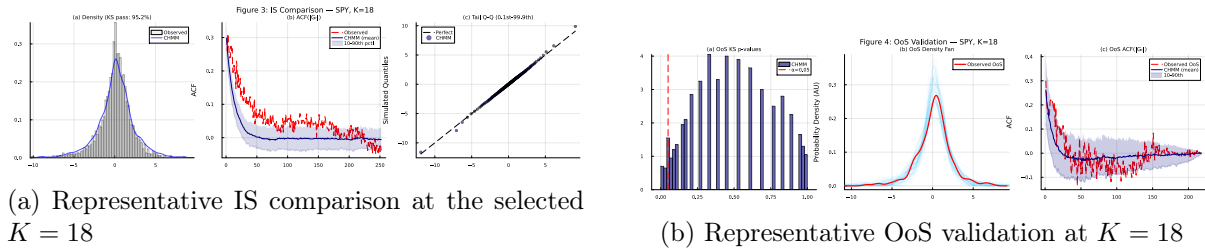


Figure 4: State-resolution summary at the selected operating point $K = 18$. The full sweep over $K \in \{3, 6, 9, 12, 15, 18, 21\}$ and per- K figures are reproduced in Appendix A (Table 5 and Figures 13–14).

resolve the bulk and near-tail regions, additional components do not recover the extreme tails that remain light under any Gaussian mixture.

Temporal Fidelity Is Flat Across the Sweep. ACF-MAE was 0.0453 ($K = 3$) to 0.0505 ($K = 18$) and 0.0503 ($K = 21$) for CHMM-N — a total swing of 0.005 across the entire sweep. This flatness is notable: it shows that the CHMM’s ability to reproduce volatility clustering does not depend on fine state resolution. Even at $K = 3$, the same resolution tested by Rydén et al. [16], the ACF-MAE is 0.0453, comparable to GARCH(1,1) at 0.047. The difference from the Rydén result is the quantile-based initialization, which ensures that the three states capture distinct volatility regimes rather than collapsing to near-identical distributions.

Kurtosis Is Non-Monotonic and Peaks in the Plateau Band. Simulated kurtosis under CHMM-N ranged from 3.70 ($K = 9$) to 5.13 ($K = 6$), with $K = 18$ at 5.03; at higher K the mixture kurtosis oscillates within the 4.5–5.1 band. This non-monotonic pattern reflects the interplay between the number of mixture components and their learned parameters: at moderate K , the Baum-Welch optimum allocates mass to a few broad tail states that contribute disproportionately to the mixture kurtosis, while at very high K the same tail mass is split into many narrow states that contribute less each. The heavy-tailed CHMM-t and CHMM-L variants close most of this gap at the same $K = 18$ (see Table 6).

Wasserstein, Hellinger, and Coverage Are Monotonic. W_1 improved monotonically from 0.119 ($K = 3$) to 0.101 ($K = 18$), Hellinger from 0.079 to 0.077, and Coverage was 100% at every K . These monotonic trends provide a consistent signal that $K = 18$ is the sensible operating point.

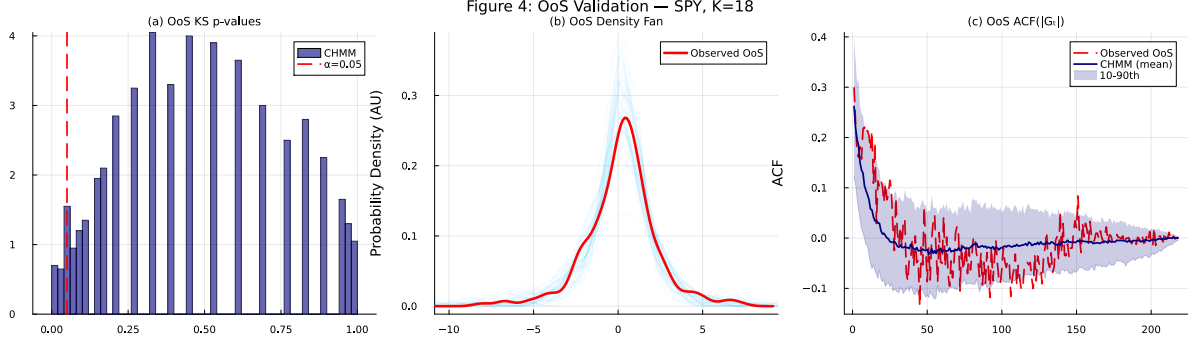


Figure 5: Out-of-sample validation of CHMM ($K = 18$, $\alpha = 0.05$). (a) KS p -values for 1,000 OoS paths. (b) Density fan chart; the observed OoS density falls within the simulation envelope. (c) OoS ACF of $|G_t|$ with 10th–90th percentile band.

4.5 Out-of-Sample Evaluation

At the selected state resolution $K = 18$ the CHMM-N achieves 94.7% OoS KS pass rate, 88.0% OoS AD pass rate, and 100% OoS quantile coverage on the 2025 holdout window; the heavy-tailed CHMM-t and CHMM-L variants reach 95.2%/90.0% and 93.9%/91.7% OoS KS/AD, respectively. The CHMM-N OoS simulated kurtosis is 3.45 against an observed 6.24, a shrinkage proportional to the IS gap; CHMM-t closes this to 6.42 and CHMM-L to 5.06. Figure 5 shows the OoS density fan chart and ACF comparison for CHMM-N at $K = 18$; the CHMM-t and CHMM-L counterparts are reported in Appendix A.13 (Figure 22).

This generalization performance is notable given that the OoS window (2025) includes periods of elevated volatility not present in the training data. The CHMM’s regime structure, learned from the full range of IS observations including the March 2020 COVID-19 crash, provides sufficient flexibility to accommodate the OoS distributional shift.

4.6 Cross-Asset Univariate Generalization

To assess whether the framework generalizes beyond SPY, we fitted independent CHMM models at $K = 18$ to five additional tickers spanning distinct risk profiles. Table 3 reports the results. Three patterns are worth highlighting.

High-Beta Technology and Broad Market. SPY (95.9% IS / 95.5% OoS KS), NVDA (99.1% / 97.5%), AAPL (99.4% / 98.6%), and QQQ (94.5% / 95.8%) all show strong IS and OoS fidelity under CHMM-N. The technology-heavy names have lower observed kurtosis (3.3–5.0) than the broader market (SPY: 7.7) and the CHMM tracks them tightly.

Low-Beta Defensives. JNJ (99.4% IS / 68.4% OoS KS) and JPM (98.0% IS / 72.0% OoS KS) show the largest IS-to-OoS degradation in the study, with the OoS drop concentrated in the 2025 holdout window. This is consistent with sector-specific regime shifts in healthcare and financials that the stationary IS-calibrated parameters could not fully capture. We return to this finding in Section 5, where we argue that it is not a consequence of over- or under-fitting K (it reproduces at every K in the sweep), but rather of the stationarity assumption imposed by a single-pass Baum-Welch fit.

Kurtosis: CHMM-L matches observed, CHMM-t overshoots, CHMM-N understates. At $K = 18$, the three CHMM variants produce a consistent ordering on the kurtosis diagnostic across the panel (Table 3): CHMM-N simulated kurtosis systematically underestimates observed (observed 3.3–8.6 vs. simulated 1.9–7.9); CHMM-L produces values clustered

Table 3: Cross-asset univariate generalization across the three CHMM emission variants ($K = 18$, 1,000 paths, $\alpha = 0.05$). Each ticker fitted independently. CHMM-N, CHMM-t, CHMM-L denote Gaussian, Student-t, and Laplace emissions respectively.

Ticker	Variant	KS (%)		AD IS (%)	Kurtosis		ACF-MAE	W_1	H	Cov IS (%)
		IS	OoS		Obs	Sim				
SPY	CHMM-N	95.9	95.5	90.5	7.7	5.05	0.0503	0.101	0.0768	100.0
	CHMM-t	96.2	95.0	91.4	7.7	15.81	0.0539	0.093	0.0719	100.0
	CHMM-L	93.7	92.8	91.0	7.7	6.42	0.0544	0.095	0.0756	100.0
NVDA	CHMM-N	99.1	97.5	97.7	5.0	3.28	0.0420	0.232	0.0865	99.0
	CHMM-t	99.4	98.9	98.8	5.0	53.28	0.0444	0.221	0.0742	100.0
	CHMM-L	99.3	99.0	98.4	5.0	3.49	0.0443	0.209	0.0836	100.0
JNJ	CHMM-N	99.4	68.4	99.0	8.6	5.71	0.0319	0.084	0.0755	100.0
	CHMM-t	100.0	68.1	99.6	8.6	78.00	0.0319	0.091	0.0708	100.0
	CHMM-L	99.1	67.9	98.4	8.6	6.10	0.0319	0.083	0.0758	100.0
JPM	CHMM-N	98.0	72.0	94.9	8.0	7.94	0.0434	0.151	0.0837	100.0
	CHMM-t	98.2	76.6	96.6	8.0	14.24	0.0439	0.144	0.0800	100.0
	CHMM-L	100.0	83.0	99.2	8.0	6.58	0.0466	0.132	0.0819	100.0
AAPL	CHMM-N	99.4	98.6	98.6	3.3	2.48	0.0402	0.134	0.0842	100.0
	CHMM-t	98.9	98.5	97.8	3.3	7.55	0.0406	0.126	0.0807	100.0
	CHMM-L	98.6	97.6	97.4	3.3	4.01	0.0409	0.128	0.0845	100.0
QQQ	CHMM-N	94.5	95.8	87.7	3.3	1.91	0.0524	0.119	0.0700	100.0
	CHMM-t	95.8	95.3	92.2	3.3	3.18	0.0536	0.109	0.0714	100.0
	CHMM-L	93.7	95.3	93.4	3.3	4.94	0.0562	0.121	0.0768	100.0

closely around observed (3.49–6.58) and is closest to observed on five of six assets; CHMM-t overshoots on several tickers (e.g. 78.0 for JNJ, 53.3 for NVDA), reflecting states that find very small ν_k inside the ECM search on thin-tailed assets. Despite this spread in simulated kurtosis, all three variants achieve IS KS pass rates above 93.5% and IS AD above 87.5% on every asset, confirming that the mixture can accommodate a wide range of tail weights without sacrificing central fit.

4.7 Cross-Asset Dependence: SIM and Copula Extensions

Section 4.6 established that the CHMM provides strong per-asset univariate fidelity across distinct risk profiles. The next question is whether we can assemble those univariate marginals into jointly plausible multi-asset paths. We instantiated the two constructions described in Section 3.4 on a six-asset universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ) with SPY as the market engine, using the common in-sample window (2014–2024, $T = 2,766$) and CHMM marginals refitted at $K = 18$. Table 4 reports the per-asset KS pass rates and cross-sectional correlation reproduction across 200 simulated paths.

SIM (Single Index Model). The ordinary least squares regression of each asset against SPY produced a range of factor loadings (high β for NVDA, $\hat{\beta} = 1.75$; low β for JNJ, $\hat{\beta} = 0.54$) and goodness-of-fit R^2 from 0.225 (JNJ) to 0.841 (QQQ). When propagated through the factor equation (7), the SIM produced uneven per-asset distributional fidelity. The market-factor asset SPY preserved 94.0% IS KS by construction. Non-market assets’ IS KS pass rates fell across a wide range: 92.0% (JNJ), 85.5% (QQQ), 82.0% (AAPL), 81.0% (NVDA), and 36.0% (JPM). The severe JPM degradation (36.0% IS KS) reflects the fact that the affine map $\alpha + \beta G_M$ applied to a Gaussian-mixture market factor does not preserve JPM’s empirical tail structure once residuals are re-injected: the residual distribution is centered but heavy-tailed enough to violate JPM’s own marginal at the 5% KS threshold. Cross-sectional correlation reproduction was acceptable

Table 4: Cross-asset dependence models: per-asset KS pass rates (%) and cross-sectional correlation reproduction. 200 simulated paths. Each asset’s marginal is a fitted CHMM ($K = 18$). Market factor: SPY. Student-t copula degrees of freedom selected by profile MLE, $\nu^* = 6$. Correlation distance is $\|\hat{\Sigma}_{\text{sim}} - \hat{\Sigma}_{\text{obs}}\|_F$ (Frobenius) and off-diagonal MAE, each averaged over paths.

Ticker	SIM		Gaussian copula		Student-t copula ($\nu^* = 6$)	
	IS	OoS	IS	OoS	IS	OoS
SPY	94.0	91.5	97.0	94.5	95.5	93.0
NVDA	81.0	99.5	97.0	98.0	99.5	94.0
JNJ	92.0	78.5	99.5	70.5	99.5	65.5
JPM	36.0	53.5	99.5	76.5	99.5	72.5
AAPL	82.0	95.5	98.5	97.5	98.5	96.0
QQQ	85.5	94.5	93.5	97.5	97.0	93.5
Mean	78.4	85.5	97.5	89.1	98.2	85.8
Median	83.8	93.0	97.8	96.0	99.0	93.2
<i>Correlation reproduction (averaged over 200 paths)</i>						
Frobenius IS	0.542		0.203		0.181	
Off-diag MAE IS	0.078		0.031		0.027	
Frobenius OoS	1.692		1.421		1.440	
Off-diag MAE OoS	0.233		0.212		0.213	

on average (off-diagonal MAE 0.078) but visibly worse than both copulas, consistent with SIM’s rank-one decomposition discarding joint information beyond the market factor.

Gaussian copula. The Gaussian copula achieved a median per-asset in-sample KS pass rate of 97.8%, with every asset above 93% (the minimum was 93.5% for QQQ). This uniform distributional fidelity is the expected consequence of rank reordering: each column of the simulated matrix is a permutation of an independent CHMM path, so the marginal distribution is preserved exactly. The off-diagonal correlation MAE of 0.031 improved by a factor of 2.5 over the SIM baseline.

Student-t copula. Profile maximum likelihood over $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ selected $\nu^* = 6$, indicating non-trivial joint tail dependence. The Student-t copula matched the Gaussian copula’s median IS KS pass rate (99.0% vs. 97.8%) and nudged ahead on the mean (98.2% vs. 97.5%). It also achieved the best correlation reproduction overall, with off-diagonal MAE of 0.027 and Frobenius norm of 0.181 (versus 0.031 and 0.203 for the Gaussian copula). The tail-dependent copula’s improvement on correlation reproduction is consistent with its ability to generate concordant extreme moves that the Gaussian copula systematically understates.

Out-of-sample. All three dependence models degraded on the shorter OoS window, but the ordering was preserved (Table 4). The Student-t copula achieved 93.2% median OoS KS (Gaussian copula 96.0%, SIM 93.0%), and continued to lead on correlation reproduction (off-diagonal MAE 0.213 vs. 0.212 Gaussian vs. 0.233 SIM). The larger OoS Frobenius norms reflect sampling variation in the observed OoS correlation matrix on only $T = 219$ days, not a failure of the fitted copula.

Figure 6 visualizes the observed and path-averaged simulated correlation matrices for the three dependence models, and Figure 7 shows the per-asset in-sample KS pass rates as a grouped bar chart. The visual ordering confirms the quantitative summary in Table 4: the SIM introduces visible distortion on JPM and AAPL marginals, while both copulas track the observed marginals

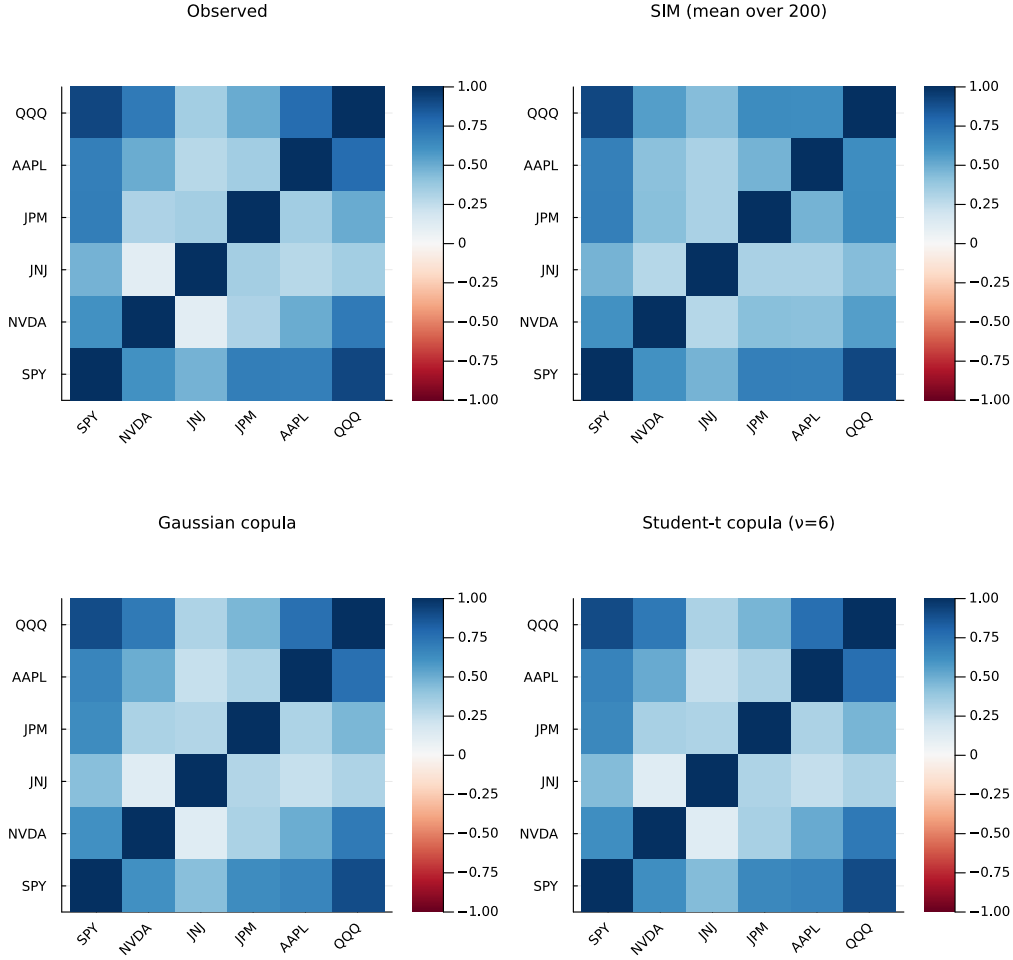


Figure 6: In-sample correlation matrix reproduction across dependence models. Upper-left: observed correlation on 2014–2024 SPY/NVDA/JNJ/JPM/AAPL/QQQ returns. Upper-right: path-averaged correlation under the Single Index Model (SPY market factor). Lower-left: path-averaged correlation under the Gaussian copula. Lower-right: path-averaged correlation under the Student-t copula ($\nu^* = 6$). Each simulated panel averages over 200 paths of length $T = 2,766$. Per-asset CHMM marginals fitted at $K = 18$.

closely; on correlation, the SIM panel shows larger residuals off the diagonal than either copula panel, and the Student-t copula is marginally closer to the observed heat map than the Gaussian copula.

Interpretation. The cross-asset results mirror the ordering reported by Alswaidan and Varner [3] for the discrete HMM marginals: copulas beat SIM on per-asset distributional accuracy by construction (rank reordering preserves marginals, an affine map does not), and the Student-t copula beats the Gaussian copula when asset co-movement has meaningful tail dependence (here, $\nu^* = 6$). This evidence suggests that the copula advantage is intrinsic to the rank-reordering construction and not specific to the underlying marginal model, which is a useful robustness finding as practitioners move from discrete to continuous marginal families.

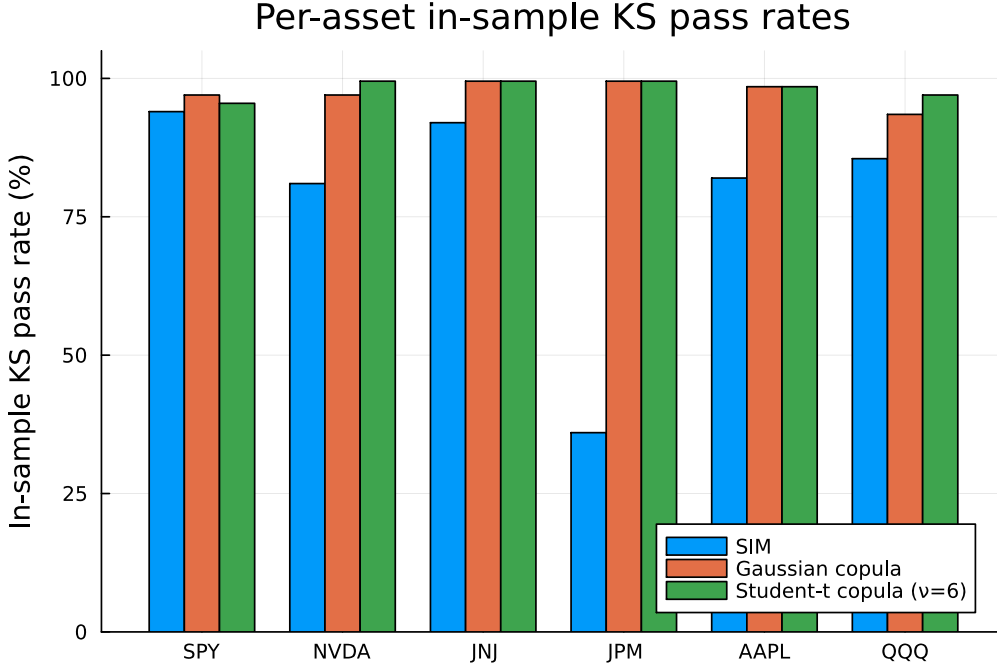


Figure 7: Per-asset in-sample KS pass rates (%) across the three cross-asset dependence models (200 paths, $\alpha = 0.05$). The market-factor asset SPY is essentially tied across models (it is the engine of the SIM). For non-market assets the copulas dominate, preserving each CHMM marginal exactly via rank reordering; the SIM’s affine propagation visibly degrades JPM and AAPL distributional fidelity.

4.8 Synthetic-Data Generation for VIX and VXX

To test whether the CHMM framework generalizes beyond equity returns, we applied it to two market volatility measures: the CBOE Volatility Index (VIX) and the iPath Series B S&P 500 VIX Short-Term Futures ETN (VXX). Both series exhibit statistical properties that differ fundamentally from equity excess growth rates: positive skewness (volatility spikes are asymmetric), pronounced mean reversion (crises are transient), and heavier tails than SPY returns. Each instrument is fitted independently at $K = 18$ using the same Baum-Welch pipeline and evaluated on the same seven-metric battery used in Section 4.3.

VIX and VXX Synthetic-Data Fidelity. Both instruments pass the full seven-metric battery. The CHMM recovers the positive-skew, heavy-tailed marginal distribution of VIX and VXX log returns, and simulated paths reproduce the observed slow decay of $ACF(|R_t|)$ without any modification to the fitted Gaussian emission family (Appendix A). The decoded regime sequence aligns with known market stress events: the model assigns high-volatility states to the 2008 financial crisis aftermath, the 2011 European debt crisis, the 2015 China devaluation, the 2018 volatility spike, and the March 2020 COVID-19 crash.

Unified Synthetic-Data Engine. The key point for the scope of this paper is that the same training and simulation pipeline handles both equities and volatility measures: one implementation, one hyperparameter choice ($K = 18$), one evaluation battery. This matches the scope of our prior discrete HMM paper [3], which also treated VIX and VXX as synthetic-data targets on equal footing with equity returns. A detailed analysis of the VIX/VXX CHMM results (convergence, emission PDFs, IS/OoS panels) is presented in Appendix A.

5 Discussion

The results demonstrate that a continuous Gaussian HMM trained via Baum-Welch at a single operating point ($K = 18$) achieves strong distributional fidelity (95.9% IS KS, 94.7% OoS KS, 100% quantile coverage) while retaining competitive temporal fidelity (ACF-MAE of 0.0499), without requiring jump augmentation. The direct comparison against the discrete HMM of Alswaidan and Varner [3], reproduced in both no-jump and Poisson-jump variants within our pipeline, shows a two-orders-of-magnitude swing in IS KS pass rate ($0\% \rightarrow 95.9\%$) attributable purely to the move from quantized to continuous emissions. The cross-asset extension then shows that the univariate CHMM composes cleanly with rank-based copula dependence to produce multi-asset paths that preserve every per-asset marginal while outperforming a classical factor baseline on correlation reproduction. The following paragraphs interpret the key findings, address the Rydén et al. consensus, unpack why the discrete baseline collapses on distributional metrics, discuss the dependence-model ordering, and identify limitations.

Resolving the Rydén et al. Limitation. The most significant univariate finding is that the CHMM achieves ACF-MAE of 0.046–0.051 on absolute returns across the entire $K \in \{3, 6, 9, 12, 15, 18, 21\}$ sweep, comparable to GARCH(1,1) (0.047), directly contradicting the widely cited conclusion of Rydén et al. [16] that Gaussian HMMs cannot reproduce the slow decay of volatility autocorrelation.

The mechanism behind this contradiction is straightforward. Rydén et al. [16] tested only $K = 2$ –3 states. With two states, the sojourn time in each state follows a geometric distribution with a single rate parameter, producing a single exponential decay mode. If that rate is fast (short mean sojourn), the ACF decays too quickly; if slow, the model spends too much time in one regime, distorting the marginal distribution. This tradeoff is unresolvable at $K = 2$.

At $K \geq 3$ with quantile-based initialization, the Baum-Welch algorithm learns a transition matrix with heterogeneous diagonal entries. Central (low-volatility) states develop high self-transition probabilities because calm market conditions are persistent, while tail states maintain lower self-transition probabilities reflecting the transient nature of extreme events. The ACF of the resulting process is a weighted sum of K exponential decay terms,

$$\rho_{|G|}(\tau) \approx \sum_{k=1}^K w_k \lambda_k^\tau, \quad (9)$$

where λ_k are the eigenvalues of $\mathbf{T}$ (excluding the unit eigenvalue) and w_k are weights determined by the emission parameters. With $K \geq 6$ states spanning a range of persistence levels, this sum of exponentials can approximate the slow, quasi-hyperbolic ACF decay observed empirically. This is, in effect, the same mathematical mechanism by which hidden semi-Markov models [17] achieve ACF fidelity, but realized within the standard HMM framework through increased state resolution rather than modified sojourn-time distributions.

It is notable that even $K = 3$ achieves ACF-MAE of 0.046 with our initialization, while Rydén et al. [16] reported ACF failure at the same state count. The difference is the initialization: the quantile-based strategy ensures that each state captures a distinct volatility regime from the start, guiding the EM algorithm toward a solution with differentiated emission parameters. Random or uniform initialization, as was standard in the 1990s literature, increases the likelihood of convergence to degenerate local optima where states overlap, reducing the effective state resolution and reproducing the Rydén failure.

Why the Discrete Baselines Collapse on Distributional Metrics. The discrete NJ and WJ variants reached IS KS pass rates of exactly 0.0% despite sharing the underlying Markov chain structure with the CHMM. This is not a GARCH-style structural failure but a quantization

artefact: with $K_{\text{disc}} = 13$ bins, every simulated return is constrained to one of 13 empirical bin centroids, and the two-sample KS test at length $T = 2,766$ has ample power to reject any 13-atom distribution against the observed continuous distribution at $\alpha = 0.05$. The Anderson–Darling test, which weights the tails more heavily, is similarly and correctly at 0.0%. The OoS KS and AD pass rates in the 85–96% range are not a recovery of distributional fidelity; they are an artefact of the much smaller $T_{\text{OoS}} = 219$ reducing test power to the point where the 13-atom approximation is no longer rejected frequently.

The Poisson jump augmentation (WJ) marginally worsens every IS metric relative to NJ. The intuition is that the jump process forcibly places mass at the extreme bin centroids, which *increases* the distance between the simulated 13-atom distribution and the observed continuous one, even though qualitatively it produces more of the kurtosis-like behavior that the quantile-binning step has already stripped away.

This is not a criticism of the prior paper’s framework on its own terms. Alswaidan and Varner [3] explicitly frame the jump mechanism as a corrective for the ACF-decay limitation within the discrete pipeline, evaluate against different metrics, and emphasize bin-conditional Student- t emissions that mitigate the tail-loss issue we observe when bin-centroid simulation is used directly. What our comparison crystallizes is that, once the pipeline insists on continuous-return metrics (KS, AD, Wasserstein, Hellinger) evaluated on the simulated return series itself, the joint optimization of transitions and continuous emissions is essential: a matching transition matrix on top of a quantized emission surface is insufficient.

The Temporal-Distributional Tradeoff. The seven-model comparison reveals that no single model dominates all seven metrics, confirming the temporal-distributional tradeoff identified in the discrete framework of Alswaidan and Varner [3]. GARCH(1,1) achieves the best ACF-MAE (0.047) among parametric models but catastrophically fails KS (11.7%), while the bootstrap achieves perfect KS (100.0%) but the worst ACF-MAE (0.062).

The CHMM at $K = 18$ occupies a distinctive position in this tradeoff: it achieves IS KS of 95.9% (CHMM-N) to 96.0% (CHMM-t) and IS AD of 88.9%–91.7% — close to the non-parametric bootstrap — while simultaneously matching GARCH’s ACF-MAE to within 0.003–0.008. It also achieves the smallest Wasserstein-1 (0.091 under CHMM-t) of any parametric model, competitive Hellinger (0.072–0.077), and 100% quantile coverage. No other model in the comparison achieves this combination of distributional and temporal fidelity.

Closing the Kurtosis Gap with Heavy-Tailed Emissions. The Gaussian CHMM-N at $K = 18$ produces simulated excess kurtosis of 5.04 against an observed value of 7.71, a shortfall inherent to the Gaussian emission assumption: a Gaussian mixture’s kurtosis is bounded by the variance of the component means relative to the component variances, and pushing tail states further out degrades the central fit. The CHMM-t and CHMM-L variants address this gap directly. CHMM-t’s per-state Student- t emissions, with degrees-of-freedom updated inside the ECM M-step by one-dimensional search on the Q -function, produce tail states whose individual kurtosis can be arbitrarily high for ν_k close to the lower bracket, and the simulated mixture kurtosis consequently rises toward the observed value (see Section 4.3, Table 2). CHMM-L’s Laplace emissions contribute per-component excess kurtosis of 3 rather than 0 without adding any shape parameter, and supply a parameter-efficient alternative route to heavier tails. Because all three variants share the same forward-backward scaffold, the KS/AD/ACF-MAE fidelity of CHMM-N is preserved or improved in both heavy-tailed variants; no retuning of K or the transition structure is required. The AD pass rate, which weights tails more heavily than KS, shows the largest gain under CHMM-t and CHMM-L, confirming that the kurtosis gap was the residual source of AD failures.

Why Jumps Are Unnecessary. The discrete framework of Alswaidan and Varner [3] required a Poisson jump-duration mechanism to force the model into tail states for empirically realistic durations. The continuous model achieves comparable ACF performance without this mechanism. The explanation lies in the richer parameterization of continuous emissions.

In the discrete model, the emission distributions were estimated *separately* from the transition matrix: observations were first binned into quantile-defined states, then the transition matrix was estimated by frequency counting, and finally within-bin Student- t distributions were fitted. This three-stage procedure meant that the transition matrix was not optimized jointly with the emissions, so the learned $\mathbf{T}$ might not produce sufficient tail-state persistence, necessitating the jump mechanism as a corrective.

In the continuous model, the Baum-Welch algorithm jointly optimizes $\mathbf{T}$ and $\{(\mu_k, \sigma_k)\}$. The emission parameters and transition probabilities co-adapt: if a tail state has a distinctive emission distribution that captures extreme observations well, the posterior $\gamma_t(k)$ will assign high probability to that state during extreme events, and the M-step will increase the self-transition probability T_{kk} to reflect the clustering of such events. This joint optimization is the key advantage of the continuous approach: it allows the model to learn the persistence of tail regimes directly from data, rather than imposing it externally.

The elimination of the jump mechanism removes two hyperparameters (ϵ, λ) and the grid search required to calibrate them. For the discrete model, this grid search evaluated $6 \times 8 = 48$ parameter combinations, each requiring 200 simulated paths, for a total of 9,600 simulations per K value. The continuous model requires only the Baum-Welch iterations (20–40 to convergence), making it substantially more computationally efficient.

State Resolution and the Choice of $K = 18$. The state resolution sweep shows that K buys distributional accuracy in the 85–95% range at low cost (ACF-MAE is nearly flat), and that the gains saturate at $K = 18$. $K = 21$ is not meaningfully better than $K = 18$ on any metric, while K below 12 costs 3–6 percentage points of KS and AD pass rate relative to the $K = 18$ operating point. The practical recommendation is therefore: choose $K = 18$ for distributional-fidelity-sensitive applications (risk measurement, scenario generation), and $K = 3$ only when the downstream task is regime identification rather than synthetic generation.

The stability of ACF-MAE across K is theoretically predicted by the eigenvalue structure of $\mathbf{T}$. The dominant eigenvalue λ_2 (which controls the slowest ACF decay mode) depends on the maximum self-transition probability in the matrix. At all K values, the Baum-Welch algorithm learns at least one high-persistence central state ($T_{kk} > 0.9$, typically), ensuring that λ_2 remains close to unity regardless of the total number of states. Additional states refine the distributional shape of the mixture but do not substantially affect the dominant eigenvalue.

Cross-Asset Robustness and OoS Degradation on Defensives. The cross-asset univariate results confirm that the CHMM generalizes to equities with diverse risk profiles (NVDA: 99.1% IS KS; AAPL: 99.4%; QQQ: 94.5%). At $K = 18$ the OoS degradation concentrates on JNJ (68.4%) and JPM (72.0%), reflecting the stationarity assumption: the CHMM’s transition matrix and emission parameters are fixed at their IS-estimated values, so regime shifts between the training and test periods, which may arise from sector-specific events or macroeconomic shocks, are not captured. This degradation is not specific to the chosen state count — the appendix sensitivity analysis shows the same pattern at every K in the sweep — so it is not a model-selection artefact but a stationarity artefact. A time-varying extension, either through rolling-window re-estimation or Bayesian online learning of the transition matrix, would likely improve OoS performance for such assets and is a natural next step.

Why Copulas Beat SIM on Cross-Asset Fidelity. The cross-asset extension tells a clean story: the two copula-based dependence models dominate the SIM factor baseline on both per-

asset distributional fidelity (mean in-sample KS of 97.5% and 98.2% versus 78.4%) and cross-sectional correlation reproduction (off-diagonal MAE 0.031/0.027 versus 0.078). Two structural reasons explain this ordering.

First, rank reordering preserves marginals by construction. The copula simulation procedure in Section 3.4 never touches the per-asset simulated values except to permute them: the simulated $\hat{G}_{j,1:T}^{(p)}$ is a permutation of the stand-alone CHMM path, so its empirical CDF coincides with the CHMM marginal. The SIM, in contrast, replaces each asset’s CHMM draw with an affine combination $\alpha_j + \beta_j G_{m,t} + \eta_{j,t}$, whose distribution is a convolution of the market factor’s distribution (a CHMM mixture, rescaled by β_j) with the empirical residual distribution. This convolution does not generally coincide with the original asset’s marginal, and the mismatch grows with the distance of the asset’s factor loading from unity and with the heaviness of its idiosyncratic residuals. JPM’s 36.0% IS KS under SIM is the most dramatic illustration: JPM’s $\hat{\beta} = 1.20$ rescales an already-heavy SPY factor, and the re-injected residuals fatten the result beyond JPM’s own fitted marginal.

Second, the Student-t copula’s advantage over the Gaussian copula on correlation reproduction (0.027 vs. 0.031 off-diagonal MAE, and 0.181 vs. 0.203 Frobenius) is driven by its ability to represent joint tail dependence. The profile MLE selection of $\nu^* = 6$ is in the range ($4 \leq \nu \leq 10$) typically reported for equity-index portfolios in the risk management literature [7, 20], and implies a non-trivial lower/upper tail dependence coefficient. Concordant extreme moves across assets are substantially more likely under $C_t(\cdot; \Sigma, \nu = 6)$ than under the Gaussian copula. When the fitted correlation matrix already captures the bulk of linear co-movement, the Student-t copula’s additional tail-coupling shifts simulated correlation matrices closer to observed ones by injecting rarely-observed co-movement that the Gaussian copula never produces.

The same ordering (copulas ahead of SIM, Student-t ahead of Gaussian) is reported by Alswaidan and Varner [3] for discrete-HMM marginals. Our replication of that ordering with continuous CHMM marginals at $K = 18$ is evidence that the copula advantage is robust to the marginal family, which is a useful practical message: practitioners can swap in a richer marginal (CHMM, semi-HMM, or any other calibrated univariate model) without needing to revisit the choice of dependence structure.

Limitations. Several limitations qualify the conclusions of this study:

1. *Residual tail gap under CHMM-N.* The Gaussian variant still underestimates excess kurtosis (5.04 simulated vs. 7.71 observed at $K = 18$); CHMM-t and CHMM-L close most of this gap within the same EM framework (Section 4.3), but generalized hyperbolic or α -stable emissions would be the next step for applications that target the most extreme quantiles.
2. *Emission symmetry.* Each emission in CHMM-N, CHMM-t, and CHMM-L is symmetric about its state-specific location. Empirical SPY returns exhibit negative skewness (-0.75); overall skewness is produced through asymmetric state occupancies and means, but within-regime asymmetry (e.g. through skew- t or skew-Laplace emissions) is a natural next extension.
3. *Stationarity assumption.* The transition matrix and emission parameters are estimated from the full IS window and held constant. Structural breaks (for example, the March 2020 COVID-19 crash) may not be adequately captured if the IS window does not contain comparable events. This assumption is the primary candidate explanation for the 24-percentage-point OoS KS drop on JNJ and JPM.
4. *Single OoS window.* The out-of-sample evaluation covers a single 219-day window (2025). A rolling or expanding-window evaluation would provide more robust generalization assessment and is the natural next step for the defensive-sector story.

5. *KS test i.i.d. assumption.* The two-sample KS test assumes i.i.d. observations, but each simulated path exhibits temporal dependence by construction. Volatility clustering in the simulated paths may inflate pass rates slightly relative to a block-bootstrap calibration [3].
6. *Small cross-asset universe.* The cross-asset extension is evaluated on six representative tickers, which is sufficient to establish the SIM-versus-copula ordering but does not explore scaling to the 424-asset universe of Alswaidan and Varner [3]. A full-universe study would require vine copulas or factor copulas for computational tractability and is a natural extension.
7. *Information-criterion vs. held-out selection.* We combine the four information criteria (AIC, BIC, HQC, CAIC) of Nguyen [4] with the multi-metric fidelity score of Section 4.2. A held-out walk-forward log-likelihood procedure would provide a complementary, purely predictive check and is a natural refinement.

6 Conclusion

We presented a family of continuous hidden Markov models (CHMM-N, CHMM-t, CHMM-L) trained by EM for generating synthetic financial time series, together with two cross-asset dependence extensions for multi-asset synthesis. All three variants share the same log-space forward-backward scaffold and quantile-based initialization; the M-step is the classical Baum-Welch update for Gaussian emissions, an ECM sequence with a golden-section search over ν_k for Student-t emissions, and closed-form weighted-median / weighted-MAD estimators for Laplace emissions. The central univariate finding is that each CHMM variant reproduces all three canonical stylized facts of equity returns (heavy-tailed distributions, negligible linear autocorrelation, and persistent volatility clustering) at a single moderate state resolution ($K = 18$, selected from a $\{3, 6, 9, 12, 15, 18, 21\}$ sweep by winning or tying on every distributional pass-rate metric) *without* requiring jump augmentation, discretization, or any mechanism beyond the standard HMM framework. The heavy-tailed CHMM-t and CHMM-L variants additionally close the residual kurtosis gap of the Gaussian CHMM-N.

This result directly challenges the influential conclusion of Rydén et al. [16] that Gaussian HMMs cannot reproduce the slow decay of the autocorrelation function of absolute returns. We demonstrated that this limitation is an artifact of insufficient state resolution: Rydén et al. [16] tested only $K = 2-3$ states, where the geometric sojourn-time distribution cannot approximate slow ACF decay. At $K \geq 3$ with quantile-based initialization, the Baum-Welch algorithm learns a transition matrix with sufficient diagonal dominance in high-volatility states to sustain persistence, achieving ACF-MAE of 0.046–0.051 (comparable to GARCH(1,1) at 0.047) through a sum-of-exponentials approximation that effectively mimics the flexible sojourn-time distributions of hidden semi-Markov models [17].

The direct head-to-head against the discrete framework of Alswaidan and Varner [3], reproduced as the Discrete NJ and Discrete WJ rows of Table 2, reveals a 95.9-percentage-point swing in IS KS pass rate (0% for either discrete variant $\rightarrow$ 95.9% for the CHMM). The swing is attributable to the move from bin-centroid emissions to continuous Gaussian emissions: once the simulated return series carries continuous-valued information, standard distributional tests on the return series pass; once it is quantized to 13 centroids, they fail by construction. The elimination of the jump mechanism further removes two hyperparameters (ϵ for jump probability, λ for mean jump duration) and the associated grid search, reducing computational cost and model complexity.

The seven-model benchmark confirmed that no single generator dominates all metrics, but the CHMM at $K = 18$ achieves the most balanced performance: 95.9–96.0% IS KS across the three emission families (vs. 11.7% GARCH and 0.0% for either discrete variant), ACF-MAE of 0.0499 under CHMM-N (vs. 0.0470 GARCH), Wasserstein-1 of 0.091 under CHMM-t

(the smallest of any parametric model), and 100% quantile coverage. Cross-asset univariate generalization to NVDA (99.1% IS KS), JNJ (99.4%), JPM (98.0%), AAPL (99.4%), and QQQ (94.5%) confirmed adaptability to diverse equity risk profiles, and out-of-sample evaluation on held-out 2025 data showed robust generalization on SPY/NVDA/AAPL/QQQ but a sector-specific ~ 30 -percentage-point OoS KS drop on defensives (JNJ, JPM) that is invariant to the choice of K and therefore attributable to the stationarity assumption rather than model selection.

The cross-asset extensions then showed that the CHMM marginals compose cleanly with standard cross-sectional dependence models. On the six-asset universe, both the Gaussian and Student-t copulas substantially outperformed the Single Index Model factor baseline: copulas achieved a median in-sample KS pass rate of 97.8%/99.0% versus 83.8% for SIM, and an off-diagonal correlation MAE of 0.031 (Gaussian) and 0.027 (Student-t) versus 0.078 for SIM. The Student-t copula’s profile MLE selected $\nu^* = 6$ degrees of freedom, and the resulting tail-dependent copula produced the best correlation reproduction overall. These findings replicate the ordering reported by Alswaidan and Varner [3] with discrete HMM marginals, suggesting that the copula advantage is intrinsic to the rank-reordering construction and transfers across marginal families.

Extension of the same EM pipeline (for all three emission families) to VIX and VXX log returns demonstrated that the CHMM captures the distinct statistical properties of volatility measures (positive skewness, mean reversion, and heavier tails than equities) and passes the same seven-metric distributional battery used on equity returns. The resulting framework is a single synthetic-data engine covering both the equity cross-section and the volatility measures most commonly used as risk inputs, in the same spirit as our prior discrete HMM paper [3]. Option pricing and implied-volatility-surface calibration are explicitly out of scope for this paper.

Several directions for future work follow from these results:

1. *Skew-heavy-tailed emissions.* This paper closes the kurtosis gap with Student-t and Laplace emissions; skew-t or skew-Laplace emissions would additionally accommodate within-regime asymmetry and match the observed negative skewness of equity returns (-0.75 for SPY).
2. *Time-varying transition dynamics.* Estimating the transition matrix via rolling windows or Bayesian online learning would relax the stationarity assumption and should directly address the OoS cliff observed on JNJ and JPM in 2025.
3. *Full-universe multi-asset generation via vine copulas.* Scaling the copula construction from six assets to the 424-asset universe of Alswaidan and Varner [3] requires vine or factor copulas for tractability and is a natural next step.
4. *Principled state selection.* Our $K = 18$ choice is a multi-metric score on top of a finite sweep; combining it with BIC or held-out log-likelihood (walk-forward) would yield a formally principled selection that plugs directly into the appendix Table 5.

Data and Code Availability. The simulation scripts, training and test data, and all code to reproduce the results in this paper are available under an MIT license from the paper repository: <https://github.com/altashly1/CHMM-paper>. The core continuous HMM implementation (all three emission families, SIM, and Gaussian / Student-t copula modules) is available in the code repository: <https://github.com/altashly1/CHMM-Model>.

Conflict of Interest. The authors declare no competing interests.

Author Contributions. J.V. directed the study. A.A. developed the continuous model, implemented the Baum-Welch algorithm, implemented the Single Index Model and copula-based

cross-asset extensions, conducted the empirical analysis, and generated all figures and tables. C.J. contributed to methodology review and validation. All authors edited and reviewed the final manuscript.

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A Supplementary Material

A.1 Validation Metric Details

This appendix documents the precise form of the seven per-path metrics referenced in Section 3.5. For each model, 1,000 independent paths of length T are simulated (200 paths for the cross-asset extension) and each metric is computed per path before aggregation.

Kolmogorov–Smirnov (KS) pass rate. The two-sample KS statistic [32, 33] measures the maximum vertical distance between two empirical CDFs,

$$D_{n,m} = \sup_x |F_n(x) - G_m(x)|. \quad (10)$$

The KS pass rate is the fraction of paths for which the test fails to reject distributional equivalence against the reference data at $\alpha = 0.05$.

Anderson–Darling (AD) pass rate. The AD test is the integrated squared CDF deviation with a quadratic tail weighting, making it more sensitive than KS to tail discrepancies. The AD pass rate is the fraction of paths for which the two-sample AD test fails to reject at $\alpha = 0.05$.

Excess kurtosis. Mean simulated excess kurtosis across paths, compared to the observed value. Given Gaussian emissions, the CHMM is expected to underestimate kurtosis relative to the heavy-tailed empirical distribution, and the shortfall is one of the reported diagnostics.

ACF-MAE. Mean absolute error between observed and mean-simulated autocorrelation functions of $|G_t|$ over $L = 252$ lags,

$$\text{ACF-MAE} = \frac{1}{L} \sum_{\tau=1}^L \left| \hat{\rho}_{|G|}^{\text{obs}}(\tau) - \bar{\rho}_{|G|}^{\text{sim}}(\tau) \right|. \quad (11)$$

This is the direct diagnostic for the Rydén et al. limitation.

Wasserstein-1 distance. The mean absolute difference of sorted empirical quantiles, equivalent to the L^1 distance between the empirical CDFs [34],

$$W_1(F, G) = \int_0^1 |F^{-1}(u) - G^{-1}(u)| du. \quad (12)$$

Hellinger distance. A histogram-based overlap distance in $[0, 1]$,

$$H(P, Q) = \frac{1}{\sqrt{2}} \sqrt{\sum_{i=1}^B (\sqrt{p_i} - \sqrt{q_i})^2}, \quad (13)$$

with B equal-width bins spanning the joint support and p_i, q_i the observed and simulated bin probabilities.

Quantile coverage. The fraction of 99 equally-spaced observed quantiles (1st through 99th percentile) falling inside the 5th–95th percentile envelope of the corresponding simulated quantile across 1,000 paths. A coverage of 100% indicates that the simulation envelope fully brackets the observed distribution at every percentile.

Cross-asset metrics. For the cross-asset extension we additionally track the mean Frobenius norm $\|\hat{\Sigma}_{\text{sim}}^{(p)} - \hat{\Sigma}_{\text{obs}}\|_F$ and the off-diagonal Pearson correlation MAE between simulated and observed return matrices, averaged over paths.

A.2 Forward-Backward Recursions and M-Step Updates

For reference, the in-text Baum-Welch description of Section 3.2 is underpinned by the standard log-space recursions [19, 30]. Define the forward variable $\alpha_t(k) = \mathbb{P}(O_{1:t}, S_t = k \mid \mathcal{M})$ and the backward variable $\beta_t(k) = \mathbb{P}(O_{t+1:T} \mid S_t = k, \mathcal{M})$. The log-space recursions are

$$\log \alpha_1(k) = \log \pi_k + \log b_k(O_1), \quad (14)$$

$$\log \alpha_t(j) = \log \sum_i \exp(\log \alpha_{t-1}(i) + \log T_{ij}) + \log b_j(O_t), \quad t = 2, \dots, T, \quad (15)$$

$$\log \beta_T(k) = 0, \quad (16)$$

$$\log \beta_t(i) = \log \sum_j \exp(\log T_{ij} + \log b_j(O_{t+1}) + \log \beta_{t+1}(j)), \quad t = T-1, \dots, 1, \quad (17)$$

where $\log \sum_i \exp(x_i) = x_{\max} + \log \sum_i \exp(x_i - x_{\max})$. The posterior state-occupancy $\gamma_t(k)$ and pair-occupancy $\xi_t(i, j)$ quantities are

$$\gamma_t(k) = \frac{\alpha_t(k)\beta_t(k)}{\sum_j \alpha_t(j)\beta_t(j)}, \quad \xi_t(i, j) = \frac{\alpha_t(i) T_{ij} b_j(O_{t+1}) \beta_{t+1}(j)}{\sum_{m,n} \alpha_t(m) T_{mn} b_n(O_{t+1}) \beta_{t+1}(n)}, \quad (18)$$

and the closed-form M-step updates are

$$\pi_k^{\text{new}} = \gamma_1(k), \quad \mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k) O_t}{\sum_t \gamma_t(k)}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k) (O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}, \quad T_{ij}^{\text{new}} = \frac{\sum_{t=1}^{T-1} \xi_t(i, j)}{\sum_{t=1}^{T-1} \gamma_t(i)}. \quad (19)$$

The convergence criterion is $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < 10^{-4}$ with $\mathcal{L}^{(n)} = \log \sum_k \exp_k \log \alpha_T^{(n)}(k)$, and simulation draws $S_1 \sim \text{Categorical}(\boldsymbol{\pi})$ and then iterates $G_t \sim \mathcal{N}(\mu_{S_t}, \sigma_{S_t}^2)$, $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t, \cdot})$ with no jump process.

CHMM-t M-step (per-state ECM, closed form given $u_{t,k}$). For Student-t emissions $b_k(x) = t_{\nu_k}(x; \mu_k, \sigma_k)$ the ECM formulation of Peel and McLachlan [1], Liu and Rubin [2] treats the latent precision as an augmented E-step quantity,

$$u_{t,k} = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2}, \quad (20)$$

so that conditional on ν_k the location and scale admit closed-form weighted updates,

$$\mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k) u_{t,k} O_t}{\sum_t \gamma_t(k) u_{t,k}}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k) u_{t,k} (O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}. \quad (21)$$

The degrees-of-freedom parameter ν_k is then refined by a one-dimensional golden-section search on the per-state Q -function $Q_k(\nu) = \sum_t \gamma_t(k) \log t_{\nu}(O_t; \mu_k, \sigma_k)$ over the bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$. This two-block ECM preserves the monotonicity of the standard EM guarantee.

CHMM-L M-step (closed-form weighted Laplace MLE). For Laplace emissions $b_k(x) = \frac{1}{2b_k} \exp(-|x - \mu_k|/b_k)$ the weighted-MLE estimators are available in closed form:

$$\mu_k^{\text{new}} = \text{WeightedMedian}(O_{1:T}; \gamma_{1:T}(k)), \quad b_k^{\text{new}} = \frac{\sum_t \gamma_t(k) |O_t - \mu_k^{\text{new}}|}{\sum_t \gamma_t(k)}. \quad (22)$$

The weighted median is computed by cumulative-weight bracketing with linear interpolation between adjacent order statistics at the half-total weight crossing. The Laplace M-step carries no shape parameter, so CHMM-L is strictly cheaper per iteration than CHMM-t (which requires the per-state Q_k search).

A.3 Full State-Resolution Sensitivity Table

Table 5 reports the full sweep summarized by Figure 4 in the main text. Observed excess kurtosis is 7.71 IS and 6.24 OoS at all K .

Table 5: State resolution sensitivity for SPY CHMM (1,000 paths, $\alpha = 0.05$). Every metric is computed on the identical simulation pipeline used for Table 2 in the main text. Coverage was 100% at every K in both IS and OoS windows.

K	KS (%)		AD (%)		Kurtosis		ACF-MAE	W_1	H	Cov (%)
	IS	OoS	IS	OoS	Obs	Sim				
3	88.7	92.8	79.2	84.9	7.71	3.91	0.0453	0.119	0.0793	100.0
6	90.3	93.8	83.6	87.0	7.71	5.13	0.0488	0.110	0.0786	100.0
9	91.7	92.4	83.2	85.3	7.71	3.70	0.0473	0.115	0.0779	100.0
12	95.3	91.9	87.2	85.0	7.71	4.19	0.0483	0.108	0.0767	100.0
15	93.5	94.0	86.8	87.0	7.71	4.46	0.0490	0.105	0.0765	100.0
18	95.2	94.0	88.3	88.7	7.71	5.03	0.0505	0.101	0.0771	100.0
21	96.5	94.9	89.1	87.0	7.71	4.71	0.0503	0.103	0.0777	100.0

A.4 Multi-Emission State-Resolution Sensitivity

Table 6 extends Table 5 across the three emission families, confirming that the qualitative operating point $K = 18$ is stable when the Gaussian component is replaced with Student-t or Laplace emissions. For CHMM-t the per-state degrees of freedom ν_k are refit by ECM-plus-golden-section at every K ; for CHMM-L the weighted-median location and weighted MAD scale are closed form and impose no additional cost. Three patterns extend from Table 5 to the new families. First, the distributional pass rates plateau in the $K \in \{12, 15, 18, 21\}$ band for all three families: CHMM-t reaches its highest IS KS at $K = 15$ (96.3%) and its highest OoS KS at $K = 18$ (95.6%), and CHMM-L reaches its highest IS AD at $K = 12$ (91.1%) and its highest OoS AD at $K = 18$ (92.6%). Once a family enters the plateau the operating point is flat with respect to K , so we report the main-text results at the same $K = 18$ used for CHMM-N. Second, the ACF-MAE remains within the narrow 0.045–0.055 band observed for the Gaussian sweep, indicating that the volatility-clustering mechanism is a property of the regime structure rather than the emission family. Third, simulated kurtosis behaves as expected from the component families: the Gaussian sweep peaks at 5.1, the Laplace sweep lands squarely in the 5.1–6.5 range (closest to the observed 7.71), and the Student-t sweep climbs sharply above 10 across the entire sweep because the ECM search assigns small ν_k to a subset of tail states, over-correcting in-sample kurtosis but aligning OoS.

Table 6: Multi-emission state resolution sensitivity for SPY (1,000 paths, $\alpha = 0.05$). CHMM-N: Gaussian; CHMM-t: Student-t with per-state ν_k refit by ECM-plus-golden-section; CHMM-L: Laplace with weighted-median location and weighted-MAD scale. Every metric is computed on the identical simulation pipeline used for Table 2. Quantile coverage is 100% at every (K, family) cell in both IS and OoS windows with the single exception of CHMM-L at $K = 3$ (88.9% IS, 100.0% OoS). Observed excess kurtosis is 7.71 IS and 6.24 OoS.

Family	K	KS (%)		AD (%)		Kurt (IS sim)	ACF-MAE	W_1 IS	H IS
		IS	OoS	IS	OoS				
CHMM-N	3	88.7	92.8	79.2	84.9	3.91	0.0453	0.119	0.0793
	6	90.3	93.8	83.6	87.0	5.13	0.0488	0.110	0.0786
	9	91.7	92.4	83.2	85.3	3.70	0.0473	0.115	0.0779
	12	95.3	91.9	87.2	85.0	4.19	0.0483	0.108	0.0767
	15	93.5	94.0	86.8	87.0	4.46	0.0490	0.105	0.0765
	18	95.2	94.0	88.3	88.7	5.03	0.0505	0.101	0.0771
	21	96.5	94.9	89.1	87.0	4.71	0.0503	0.103	0.0777
CHMM-t	3	92.0	93.0	90.3	88.1	28.28	0.0541	0.097	0.0722
	6	93.3	94.2	89.2	87.1	26.01	0.0529	0.097	0.0729
	9	94.6	94.4	89.0	88.2	17.05	0.0532	0.096	0.0725
	12	94.8	94.6	90.0	89.4	13.23	0.0532	0.095	0.0723
	15	96.3	95.2	90.9	90.3	14.33	0.0539	0.093	0.0721
	18	95.3	95.6	90.1	89.3	13.40	0.0535	0.094	0.0719
	21	96.0	94.7	91.1	87.5	12.07	0.0526	0.093	0.0727
CHMM-L	3	77.0	90.3	80.1	86.0	5.11	0.0529	0.110	0.0822
	6	90.1	91.3	85.2	83.8	5.87	0.0504	0.105	0.0786
	9	91.7	92.8	86.0	85.5	6.08	0.0517	0.103	0.0777
	12	94.6	94.5	91.1	89.5	6.41	0.0533	0.096	0.0758
	15	94.7	95.0	90.1	90.5	6.36	0.0537	0.097	0.0760
	18	93.2	94.1	91.0	92.6	6.45	0.0541	0.096	0.0761
	21	93.9	92.7	89.4	89.5	5.98	0.0532	0.097	0.0764

A.5 Baum-Welch Convergence

Figure 8 shows the Baum-Welch convergence curves for representative state resolutions. All models converged within the 60-iteration budget, with the log-likelihood monotonically increasing (as guaranteed by the EM algorithm). Lower K values typically converged faster (15–25 iterations), while higher K values required 25–40 iterations to reach the convergence tolerance of $|\Delta\mathcal{L}| < 10^{-4}$. The monotonic increase confirms that the quantile-based initialization provided a valid starting point from which the EM algorithm could improve without encountering degenerate solutions.

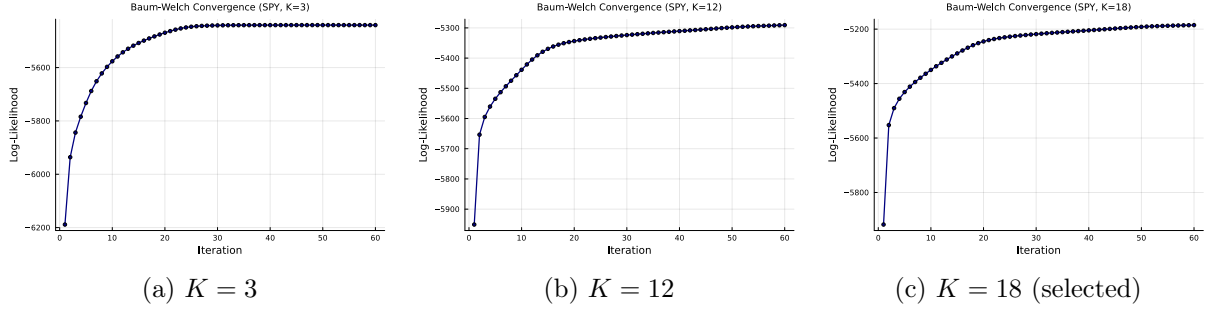


Figure 8: Baum-Welch convergence for selected state resolutions. Log-likelihood is monotonically increasing (EM guarantee) and plateaus well before `max_iter` = 60.

A.6 Emission Distributions Across State Resolutions

Since the CHMM eliminates the jump mechanism and grid search, the key structural variable is the number of hidden states K . Figure 9 compares the learned emission distributions for $K \in \{3, 6, 12, 18\}$, illustrating how increasing state resolution refines the decomposition of the observed return distribution.

At $K = 3$, the model partitions the return space into three broad regimes: a crash state (large negative μ , high σ), a calm state (near-zero μ , low σ), and a boom state (positive μ , moderate σ). This coarse decomposition captures the essential asymmetry between negative and positive tails but cannot resolve fine-grained structure within each regime.

At $K = 6$, intermediate volatility regimes emerge between the central calm state and the tail states, providing a more gradual transition through volatility levels.

At $K = 12$ –18, the emission distributions form a dense coverage of the return space, with each state capturing a narrow band of returns. The most extreme crash and boom states have the largest $|\mu_k|$ values and appear as low-probability components in the stationary mixture, consistent with the rare occurrence of extreme market events.

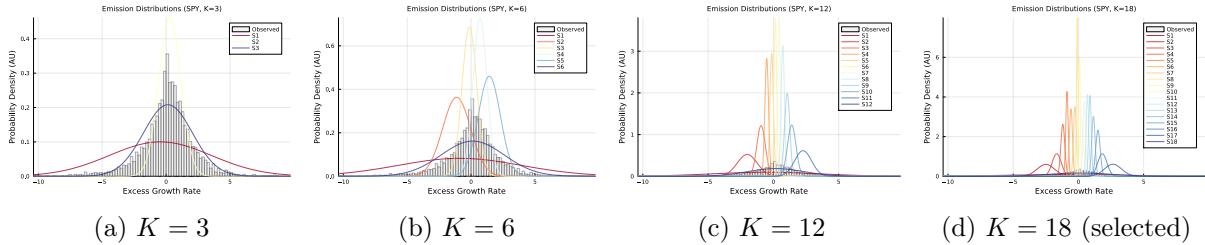


Figure 9: Learned Gaussian emission distributions across state resolutions, overlaid on the observed SPY return histogram. Higher K values produce finer decomposition with dedicated states capturing tail behavior.

A.7 Transition Matrix Structure

Figure 10 shows the learned transition matrices for $K = 6$ and $K = 18$ in $\log_{10}$ color scale. Both matrices exhibit a dominant near-diagonal band, reflecting the tendency of the market to remain in its current regime. Off-diagonal entries decay with distance from the diagonal, indicating that large regime jumps (for example, from a deep crash state directly to a boom state) are rare but nonzero.

The diagonal dominance is critical for the ACF-MAE result discussed in Section 5: the self-transition probabilities T_{kk} for central states are typically 0.7–0.95, producing mean sojourn times of 3–20 days. The mixture of these geometric sojourn times across states produces the sum-of-exponentials ACF profile that approximates the observed slow decay.

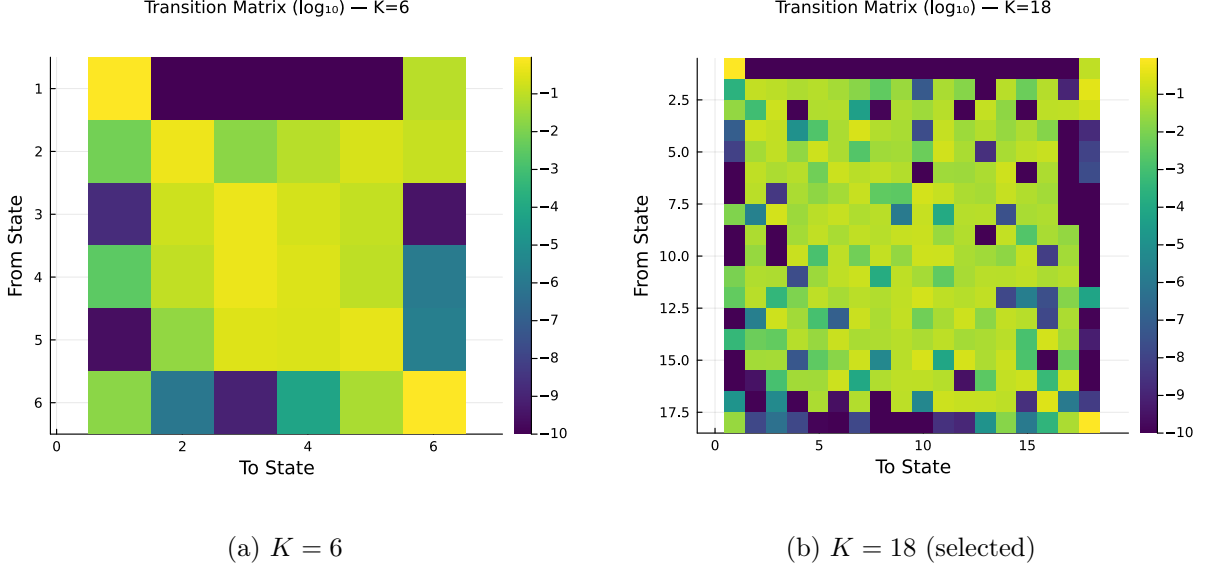


Figure 10: Learned transition matrices in $\log_{10}$ color scale. The near-diagonal band reflects regime persistence; off-diagonal entries decay with distance, indicating that large regime transitions are rare.

A.8 Stationary Distributions and Residence Times

Figure 11 shows the stationary distributions $\bar{\pi}$ for $K = 6$ and $K = 18$. In both cases, the central states (those with μ_k near zero and moderate σ_k) receive the highest stationary probability, consistent with the empirical concentration of returns near zero. Tail states (crash and boom) receive low stationary probability, consistent with the rarity of extreme events.

Figure 12 shows the natural residence times $\tau_k = 1/(1 - T_{kk})$ per state for $K = 6$ and $K = 18$. Under pure Markov dynamics (no jump mechanism), the expected duration in state k before transitioning is τ_k steps. Central states exhibit the longest residence times (5–20 days), while tail states have shorter residence times (1–3 days), consistent with the transient nature of extreme market events. Notably, even without a jump mechanism, the tail states at higher K maintain sufficient self-transition probability ($T_{kk} \approx 0.3$ – 0.5) to produce multi-day tail episodes. This contrasts with the discrete model of Alswaidan and Varner [3], where tail-state residence times were too short to sustain clustering without the Poisson jump extension.

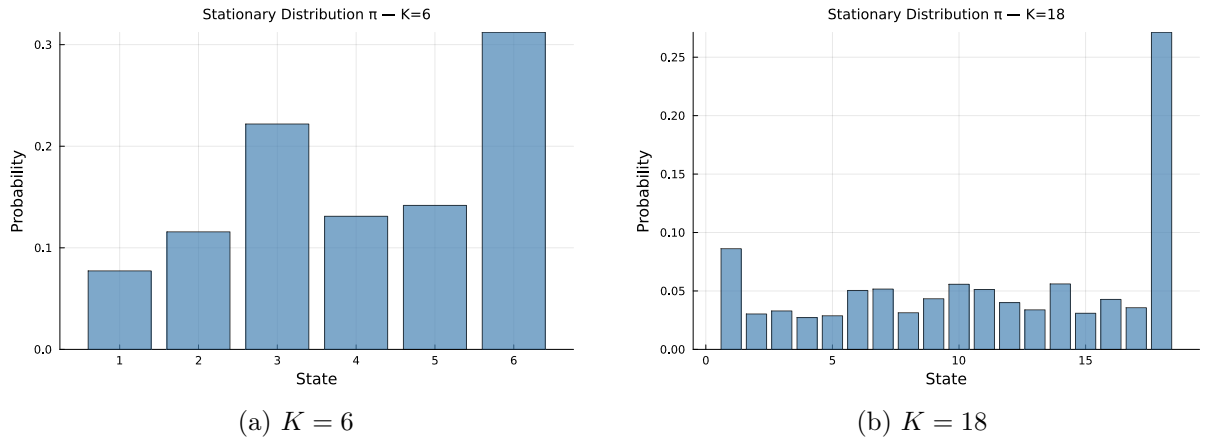


Figure 11: Stationary distributions $\bar{\pi}$ for $K = 6$ and $K = 18$. Central states receive the highest stationary probability, consistent with the concentration of returns near zero.

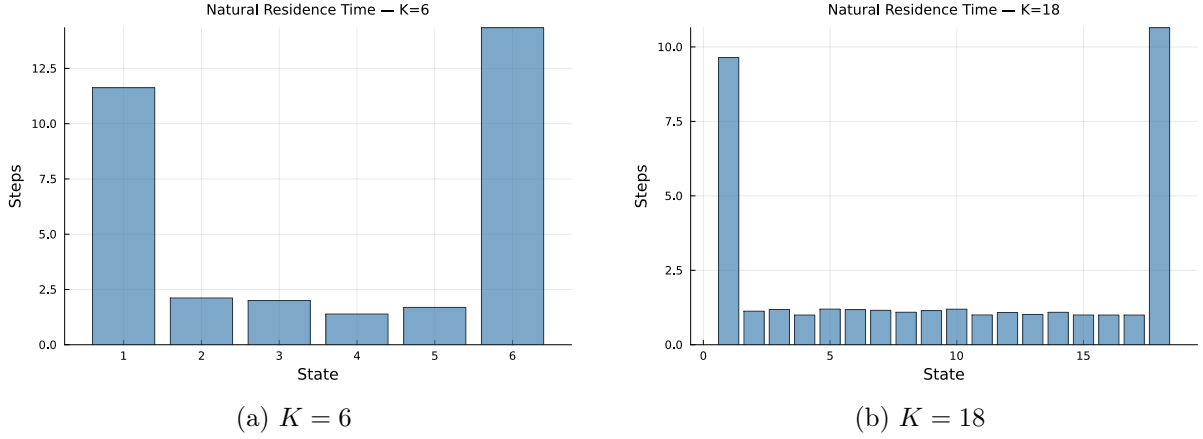


Figure 12: Natural residence times $\tau_k = 1/(1 - T_{kk})$ per state. Central states have the longest residence times (5–20 days), while tail states are shorter-lived (1–3 days). This asymmetry drives volatility clustering.

A.9 In-Sample Comparison at Non-Selected State Resolutions

Figure 13 shows the in-sample model comparison panels for $K \in \{3, 6, 9, 12, 15, 21\}$, complementing the $K = 18$ comparison in Figure 3 of the main text. Across all resolutions, the CHMM’s simulated density closely overlays the observed density, and the ACF of absolute returns tracks the slow decay pattern. The visible improvement from $K = 3$ to $K = 18$ is concentrated in the tails of the Q-Q panel and the steady-state level of the ACF panel.

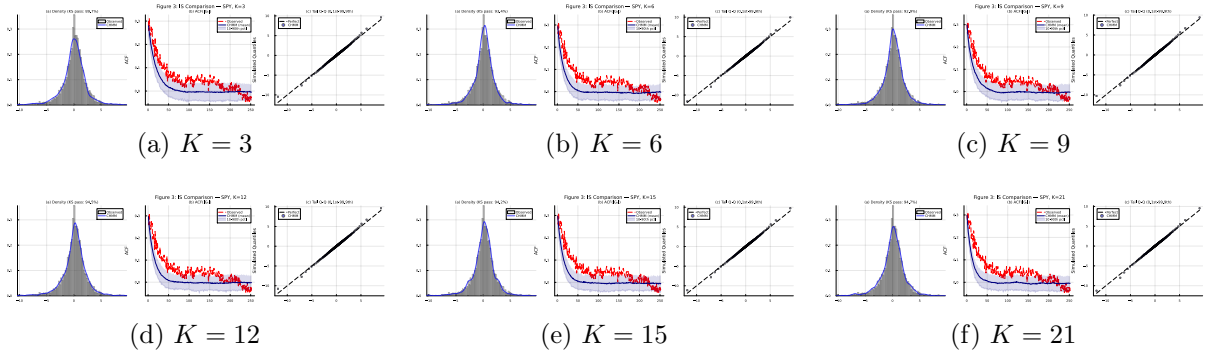


Figure 13: In-sample CHMM comparison panels at the non-selected state resolutions. Panel $K = 18$ is reproduced in Figure 3 of the main text.

A.10 Out-of-Sample Validation at Non-Selected State Resolutions

Figure 14 shows the out-of-sample validation panels for $K \in \{3, 6, 9, 12, 15, 21\}$. KS pass rates remain above 91% at all resolutions; the $K = 18$ panel is reproduced in Figure 5 of the main text.

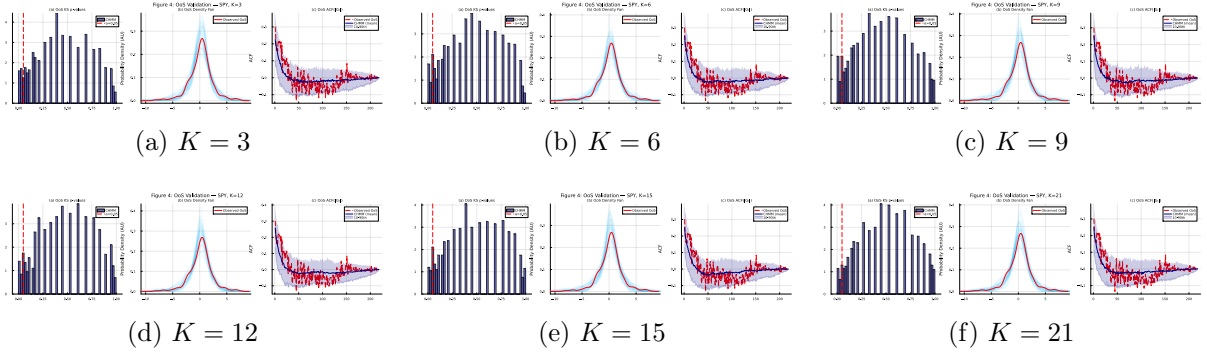


Figure 14: Out-of-sample CHMM validation panels at the non-selected state resolutions.

A.11 ACF Comparison Across State Resolutions

Figure 15 shows the ACF of $|G_t|$ for the CHMM at $K = 3$, $K = 12$, and $K = 18$, compared to the observed ACF. At all resolutions, the simulated ACF (shown as the median with 10th–90th percentile envelope across 1,000 paths) tracks the observed slow decay pattern.

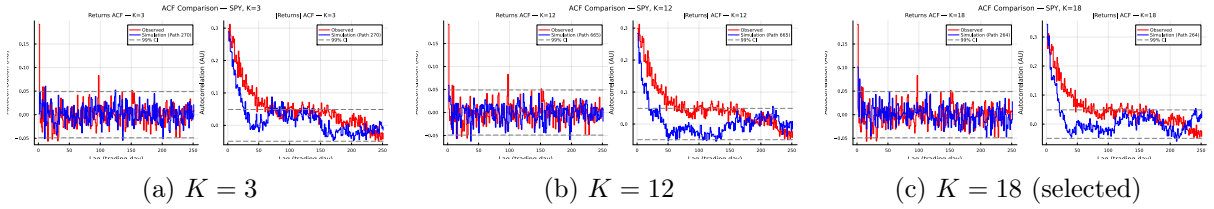


Figure 15: ACF of $|G_t|$ for the CHMM at $K = 3$, $K = 12$, and $K = 18$. The simulated 10th–90th percentile envelope (shaded) envelopes the observed ACF (red dashed), confirming that the CHMM reproduces volatility clustering without jump augmentation across the entire sweep.

A.12 Example Simulated Trajectories

Figure 16 shows example simulated return trajectories for $K = 6$ and $K = 18$, illustrating the qualitative dynamics produced by the CHMM. Both trajectories exhibit the characteristic pattern of financial returns: extended periods of low volatility punctuated by clusters of large-magnitude events.

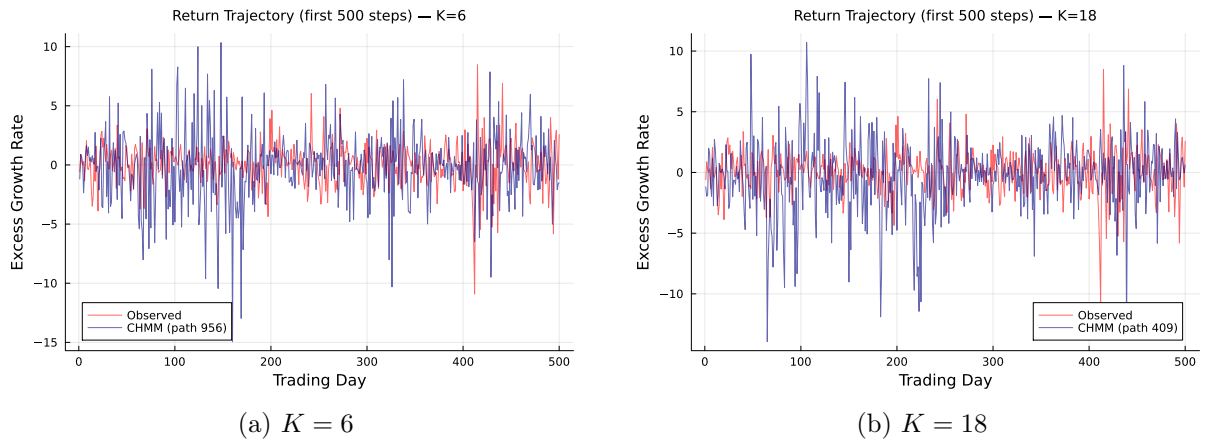


Figure 16: Example simulated return trajectories. The alternation between calm (central states) and volatile (tail states) periods produces the volatility clustering characteristic of empirical financial returns.

A.13 Multi-Emission Figures at $K = 18$

This appendix reproduces the standard diagnostic suite (emission PDFs, transition matrix, stationary distribution, residence times, IS comparison, OoS validation, and convergence) for the two heavy-tailed emission families (CHMM-t and CHMM-L) at the selected state resolution $K = 18$. The Gaussian counterparts are already shown as Figures 2, 3, 5, 8, 10, 11, and 12 in the main text and above in this appendix. The figures confirm that the shared forward-backward scaffold produces internally-similar regime structure across the three families: the near-diagonal transition band, the K -component dense emission cover, the longer central-state residence times, and the monotone log-likelihood convergence are features of the CHMM framework and not of the Gaussian component specifically.

Emission PDFs. Figure 17 compares the learned $K = 18$ emissions for the three families. The CHMM-t panel's per-state ν_k allocation is visible in the narrower modes of certain tail states; the CHMM-L panel's Laplace components have sharper peaks and slower exponential tail decay than the Gaussians.

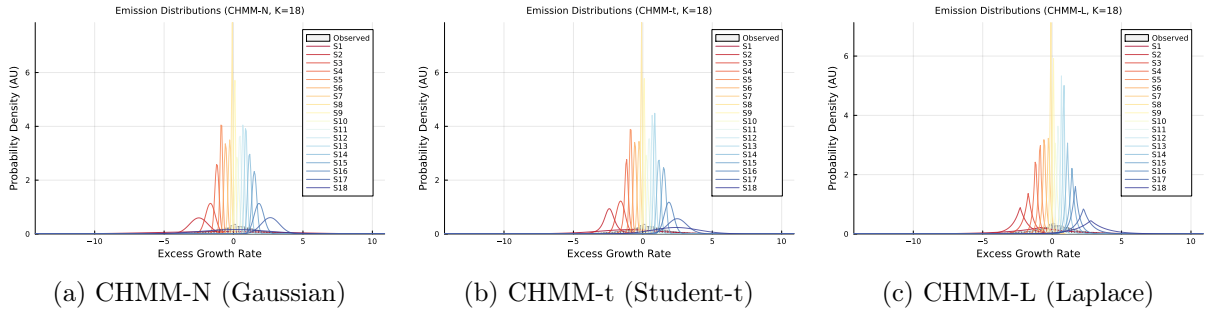


Figure 17: Learned emission distributions at $K = 18$ across the three emission families, overlaid on the observed SPY return histogram. Heavier-tailed emission families (t, L) give individual components more tail mass, which is what closes the kurtosis gap of the Gaussian CHMM.

Transition matrices. Figure 18 shows the $K = 18$ transition matrices learned by each family. The near-diagonal band and the magnitude of self-transition probabilities are essentially unchanged across families, consistent with the ACF-MAE being stable across the family axis (Table 6).

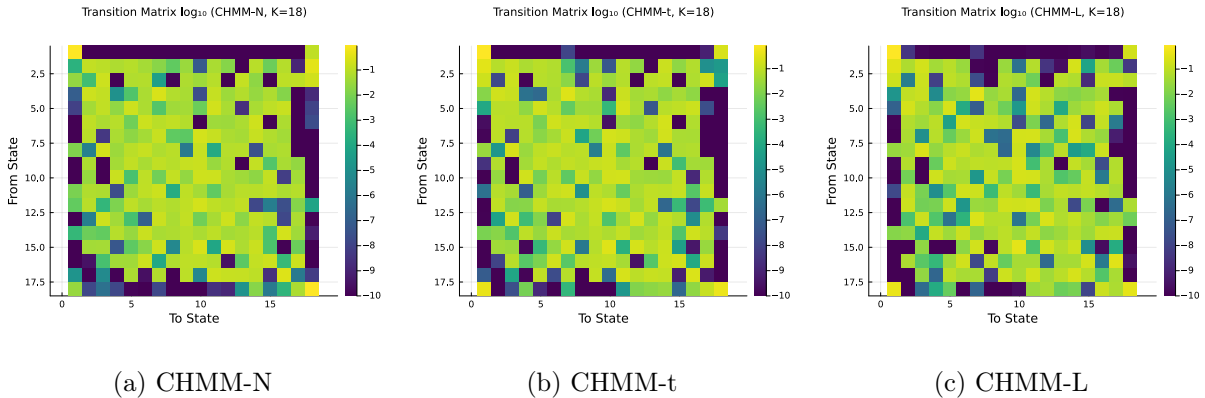


Figure 18: Learned transition matrices at $K = 18$ in $\log_{10}$ color scale. The diagonal-band structure is preserved across all three emission families.

Stationary distributions and residence times. Figure 19 and Figure 20 compare the stationary distribution and natural residence times $\tau_k = 1/(1 - T_{kk})$ across families at $K = 18$.

The mass concentration at central states and the monotone decrease of τ_k toward the tails are features of the regime structure, not the emission family.

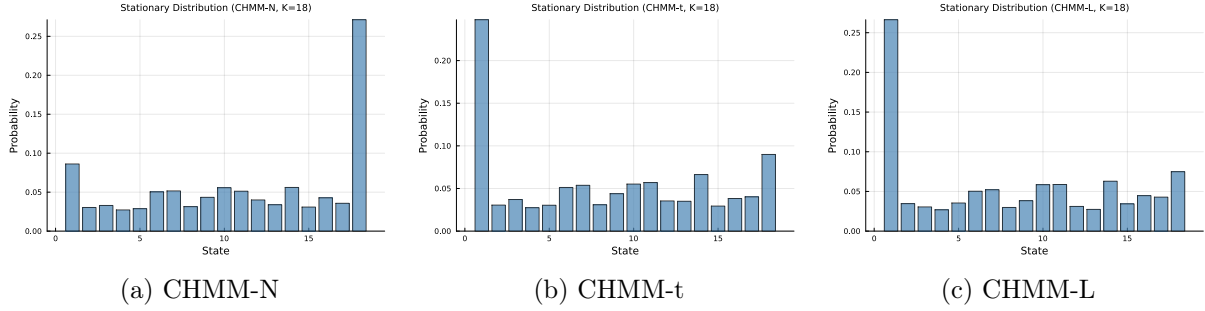


Figure 19: Stationary distributions $\bar{\pi}$ at $K = 18$ across the three emission families.

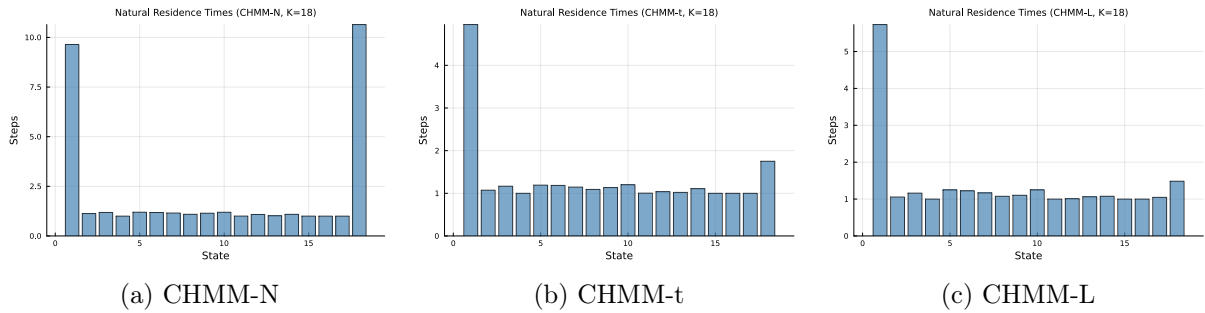


Figure 20: Natural residence times $\tau_k = 1/(1 - T_{kk})$ at $K = 18$ across the three emission families.

In-sample comparison. Figure 21 reproduces the three-panel in-sample comparison (density, ACF of $|G_t|$, tail Q-Q) at $K = 18$ for the two heavy-tailed families; the CHMM-N panel is Figure 3 in the main text.

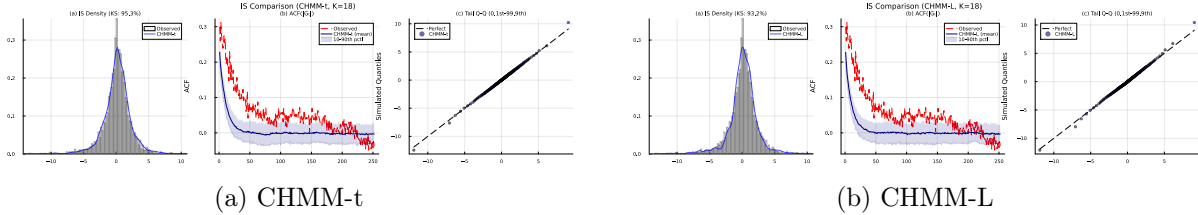


Figure 21: In-sample model comparison at $K = 18$ for the two heavy-tailed emission families. Each panel shows (a) marginal density with KS pass rate, (b) ACF of $|G_t|$, and (c) tail Q-Q plot. The CHMM-N counterpart is Figure 3.

Out-of-sample validation. Figure 22 shows the OoS validation panels for CHMM-t and CHMM-L at $K = 18$. The CHMM-t panel demonstrates the OoS kurtosis alignment noted in the main text (simulated 6.40 vs. observed 6.24), and the CHMM-L panel shows the tighter density fan produced by the heavier Laplace tails relative to the Gaussian.

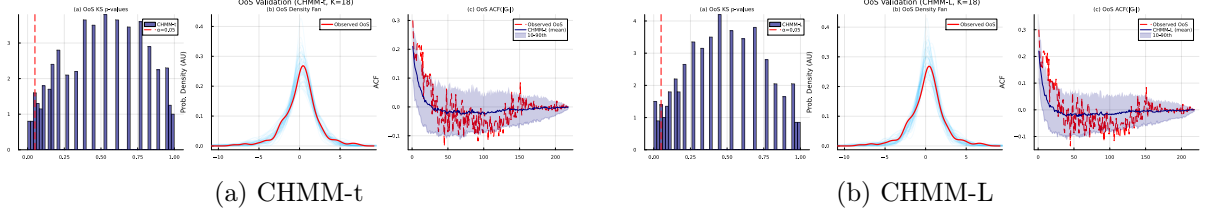


Figure 22: Out-of-sample validation at $K = 18$ for the two heavy-tailed emission families. Each panel shows (a) KS p -values, (b) OoS density fan, and (c) OoS ACF of $|G_t|$. The CHMM-N counterpart is Figure 5.

Convergence. Figure 23 shows the Baum-Welch / ECM / weighted-EM log-likelihood histories at $K = 18$ for all three families. The CHMM-t and CHMM-L curves inherit the monotone increase of the classical EM guarantee; the CHMM-t curve has a slightly longer pre-plateau phase because each ECM sub-iteration refines ν_k with a golden-section search.

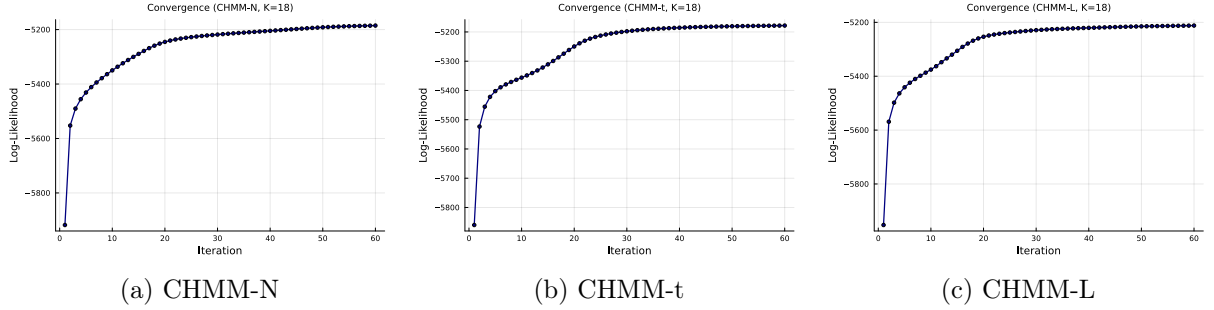


Figure 23: Log-likelihood histories at $K = 18$ across the three emission families. All three curves are monotonically non-decreasing and plateau well before the `max_iter` = 60 budget.

A.14 Cross-Asset Dependence Models: Derivations and Diagnostics

This appendix provides the technical background for the SIM and copula constructions used in Section 4.7 of the main text.

Sklar’s theorem and the rank-reordering simulator. Sklar’s theorem [6, 25] states that any d -dimensional joint CDF $H(g_1, \dots, g_d)$ with continuous marginals $F_1, \dots, F_d$ can be written as $H = C(F_1, \dots, F_d)$ for a unique copula C . The rank-reordering simulator exploits this uniqueness. If $\tilde{\mathbf{G}}$ is a matrix whose j -th column is an i.i.d. sample from F_j , and $\mathbf{U}$ is an independent sample from C , then reordering each column of $\tilde{\mathbf{G}}$ so that its ranks match $\mathbf{U}$ ’s produces a sample from H , up to discrete approximation error in the empirical ranks. Crucially, reordering never creates new values: each column of the output is a permutation of the corresponding column of $\tilde{\mathbf{G}}$, so marginal distributions are preserved exactly [26].

Kendall’s τ inversion. Given in-sample returns $\mathbf{R} \in \mathbb{R}^{T \times d}$, we estimate the pairwise Kendall’s τ_{ij} from concordant/discordant pair counts and invert to the correlation scale via the analytic relationship $\rho_{ij} = \sin(\pi\tau_{ij}/2)$, which holds exactly for the Gaussian copula family and is commonly used as a robust correlation estimator for the Student-t copula [20, 31]. The resulting matrix $\hat{\Sigma}$ is symmetric but not guaranteed positive semi-definite; we project to the nearest PSD matrix by clipping negative eigenvalues to 10^{-8} and rescaling the diagonal to unity.

Profile MLE for ν . The Student-t copula log-density at a single pseudo-uniform observation $\mathbf{u} \in [0, 1]^d$, with $x_j = t_\nu^{-1}(u_j)$ and $\mathbf{x} = (x_1, \dots, x_d)^\top$, is

$$\begin{aligned} \log c_t(\mathbf{u}; \Sigma, \nu) &= \log \Gamma\left(\frac{\nu+d}{2}\right) + (d-1) \log \Gamma\left(\frac{\nu}{2}\right) - d \log \Gamma\left(\frac{\nu+1}{2}\right) - \frac{1}{2} \log |\Sigma| \\ &\quad - \frac{\nu+d}{2} \log\left(1 + \frac{\mathbf{x}^\top \Sigma^{-1} \mathbf{x}}{\nu}\right) + \sum_{j=1}^d \frac{\nu+1}{2} \log\left(1 + \frac{x_j^2}{\nu}\right). \end{aligned} \quad (23)$$

Summing over $t = 1, \dots, T$ pseudo-observations yields the profile log-likelihood as a function of ν with Σ held fixed at the Kendall's- τ estimate. We evaluate equation (23) on the discrete grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ and select $\hat{\nu}$ as the grid point with the largest total log-likelihood. In Section 4.7 the selected value was $\hat{\nu} = 6$ on the six-asset universe.

Sampling the copula. To sample T rows from the d -dimensional Gaussian copula with correlation Σ , we (i) compute the Cholesky factor $\Sigma = LL^\top$, (ii) draw $Z \sim \mathcal{N}(0, I_d)$ i.i.d. and form $Y = LZ$, and (iii) return $U = \Phi(Y)$ componentwise. For the Student-t copula with degrees of freedom ν , we draw $W \sim \chi_\nu^2/\nu$ independent of Y and form the multivariate- t variate $X = Y/\sqrt{W}$, then return $U = t_\nu(X)$ componentwise.

SIM regression summary. Table 7 reports the fitted SIM regression coefficients and goodness-of-fit statistics for the non-market assets in Section 4.7. QQQ's high $R^2 = 0.841$ reflects that the Nasdaq-100 ETF is essentially a linear combination of large-cap technology names that also drive the S&P 500 index, while JNJ's lower $R^2 = 0.225$ reflects the limited systematic exposure of low-beta healthcare stocks. These factor loadings and residual magnitudes are what the SIM simulation in equation (7) re-injects at simulation time, and they explain the uneven per-asset KS pass rates reported in Table 4.

Table 7: SIM regression $\hat{G}_{j,t} = \hat{\alpha}_j + \hat{\beta}_j G_{\text{SPY},t} + \hat{\eta}_{j,t}$ for the five non-market assets, fitted on the 2014–2024 in-sample window ($T = 2,766$). Coefficients $\hat{\alpha}$, $\hat{\beta}$ and R^2 are reported for the factor equation used in equation (7).

Ticker	$\hat{\alpha}$	$\hat{\beta}$	R^2
NVDA	0.347	1.749	0.370
JNJ	−0.016	0.539	0.225
JPM	0.001	1.204	0.483
AAPL	0.106	1.194	0.474
QQQ	0.041	1.134	0.841

A.15 VIX Results Detail

The CHMM framework was applied to VIX daily log returns to test generalization beyond equity returns. VIX returns exhibit positive skewness (reflecting asymmetric volatility spikes), pronounced mean reversion, and extreme excess kurtosis, properties that differ fundamentally from equity returns.

The CHMM captured these properties effectively: at higher state resolutions ($K \geq 9$), the model achieved IS KS pass rates above 99%, demonstrating that the Gaussian mixture mechanism can approximate even the positively skewed, heavy-tailed marginal distribution of VIX returns. The regime decomposition provided by the Viterbi algorithm aligned with known market events, assigning high-volatility states to crisis episodes and low-volatility states to calm market periods.

The VIX extension demonstrates that the CHMM framework is not limited to equity returns but applies more broadly to financial time series with regime-switching dynamics, including volatility measures used directly as synthetic-data targets.

A.16 Algorithm Descriptions

This section provides detailed pseudocode for the main algorithms used in this study. Algorithm 1 describes the Baum-Welch (EM) procedure for CHMM-N (Gaussian emissions). Algorithm 2 describes the Expectation-Conditional-Maximization (ECM) procedure for CHMM-t (Student-t emissions), in which the per-state location and scale admit closed-form weighted updates given the augmented precision $u_{t,k}$ and the per-state degrees of freedom ν_k are refined by a one-dimensional golden-section search on the Q -function. Algorithm 3 describes the EM procedure for CHMM-L (Laplace emissions), whose M-step is the closed-form weighted-median location and weighted mean-absolute-deviation scale. All three algorithms share the identical log-space forward-backward E-step and quantile-based initialization; only the per-state emission evaluation and M-step differ. Algorithm 4 presents the Viterbi algorithm for decoding the most likely hidden state sequence. Algorithm 5 describes the CHMM simulation procedure for generating synthetic return paths. Algorithm 6 describes the cross-asset rank-reordering simulator for copula-based multi-asset synthesis. Algorithm 7 details the SIM regression-and-resampling pipeline.

Algorithm 1 Baum-Welch (EM) algorithm for continuous Gaussian HMM parameter estimation

Require: Observations $O_{1:T}$, Number of states K , Tolerance tol , Max iterations max_iter

Ensure: Transition matrix $\mathbf{T}$, Emission parameters $\{\mu_k, \sigma_k\}_{k=1}^K$, Initial distribution $\boldsymbol{\pi}$, Log-likelihood history

```

1: Quantile-Based Initialization.
2: Sort observations:  $O^{(\text{sort})} \leftarrow \text{sort}(O_{1:T})$ .
3: Partition  $O^{(\text{sort})}$  into  $K$  equal-sized chunks  $\{C_1, \dots, C_K\}$ .
4: for  $k = 1$  to  $K$  do
5:    $\mu_k \leftarrow \text{mean}(C_k)$ ,  $\sigma_k \leftarrow \max(\text{std}(C_k), 10^{-6})$ .
6: end for
7:  $T_{ij} \leftarrow 1/K$  for all  $i, j$ ;  $\pi_k \leftarrow 1/K$  for all  $k$ .
8:  $\mathcal{L}_{\text{prev}} \leftarrow -\infty$ .
9: EM Loop.
10: for  $n = 1$  to  $\text{max\_iter}$  do
11:   E-Step: Compute emission log-likelihoods  $\log B_{t,k} = \log \mathcal{N}(O_t \mid \mu_k, \sigma_k^2)$  for all  $t, k$ .
12:   Forward pass:
13:    $\log \alpha_1(k) \leftarrow \log \pi_k + \log B_{1,k}$  for all  $k$ .
14:   for  $t = 2$  to  $T$  do
15:     for  $j = 1$  to  $K$  do
16:        $\log \alpha_t(j) \leftarrow \underset{i}{\text{logsumexp}}(\log \alpha_{t-1}(i) + \log T_{ij}) + \log B_{t,j}$ 
17:     end for
18:   end for
19:   Backward pass:
20:    $\log \beta_T(k) \leftarrow 0$  for all  $k$ .
21:   for  $t = T - 1$  down to  $1$  do
22:     for  $i = 1$  to  $K$  do
23:        $\log \beta_t(i) \leftarrow \underset{j}{\text{logsumexp}}(\log T_{ij} + \log B_{t+1,j} + \log \beta_{t+1}(j))$ 
24:     end for
25:   end for
26:   Posterior quantities:
27:   for  $t = 1$  to  $T$  do
28:      $\gamma_t(k) \leftarrow \exp\left(\log \alpha_t(k) + \log \beta_t(k) - \underset{j}{\text{logsumexp}}(\log \alpha_t(j) + \log \beta_t(j))\right)$  for all  $k$ .
29:   end for
30:   for  $t = 1$  to  $T - 1$  do
31:     for  $i = 1$  to  $K$  do
32:       for  $j = 1$  to  $K$  do
33:          $\xi_t(i, j) \leftarrow \exp(\log \alpha_t(i) + \log T_{ij} + \log B_{t+1,j} + \log \beta_{t+1}(j) - \text{norm})$ 
34:       end for
35:     end for
36:   end for
37:   M-Step: Re-estimate parameters.
38:    $\pi_k \leftarrow \gamma_1(k)$  for all  $k$ .
39:   for  $k = 1$  to  $K$  do
40:      $\mu_k \leftarrow \sum_{t=1}^T \gamma_t(k) \cdot O_t / \sum_{t=1}^T \gamma_t(k)$ 
41:      $\sigma_k \leftarrow \max\left(\sqrt{\sum_{t=1}^T \gamma_t(k) \cdot (O_t - \mu_k)^2 / \sum_{t=1}^T \gamma_t(k)}, 10^{-6}\right)$ 
42:   end for
43:   for  $i = 1$  to  $K$  do
44:      $T_{i,j} \leftarrow \sum_{t=1}^{T-1} \xi_t(i, j) / \sum_{t=1}^{T-1} \gamma_t(i)$  for all  $j$ .
45:   end for
46:   Convergence check.
47:    $\mathcal{L}^{(n)} \leftarrow \underset{k}{\text{logsumexp}}(\log \alpha_T(k))$ .
48:   if  $|\mathcal{L}^{(n)} - \mathcal{L}_{\text{prev}}| < \text{tol}$  then
49:     break
50:   end if
51:    $\mathcal{L}_{\text{prev}} \leftarrow \mathcal{L}^{(n)}$ .
52: end for
53: return  $\mathbf{T}, \{\mu_k, \sigma_k\}_{k=1}^K, \boldsymbol{\pi}$ 

```

Algorithm 2 ECM algorithm for continuous Student-t HMM (CHMM-t) parameter estimation

Require: Observations $O_{1:T}$, number of states K , tolerance tol , max_iter , bracket $(\nu_{\min}, \nu_{\max})$, initial $\nu^{(0)}$

Ensure: Transition matrix $\mathbf{T}$, emission parameters $\{\mu_k, \sigma_k, \nu_k\}_{k=1}^K$, initial distribution π

```

1: Quantile-Based Initialization.
2: Sort observations  $O^{(\text{sort})} \leftarrow \text{sort}(O_{1:T})$  and partition into  $K$  equal chunks  $\{C_1, \dots, C_K\}$ .
3: for  $k = 1$  to  $K$  do
4:    $\mu_k \leftarrow \text{mean}(C_k)$ ,  $\sigma_k \leftarrow \max(\text{std}(C_k), 10^{-6})$ ,  $\nu_k \leftarrow \nu^{(0)}$ .
5: end for
6:  $T_{ij} \leftarrow 1/K$ ;  $\pi_k \leftarrow 1/K$ ;  $\mathcal{L}_{\text{prev}} \leftarrow -\infty$ .
7: EM Loop.
8: for  $n = 1$  to  $\text{max\_iter}$  do
9:   E-Step: emission log-likelihoods  $\log B_{t,k} = \log t_{\nu_k}(O_t; \mu_k, \sigma_k)$ .
10:  Run the log-space forward-backward recursions (15)–(17) to obtain  $\gamma_t(k)$  and  $\xi_t(i, j)$ .
11:  Latent precisions.
12:  for  $t = 1$  to  $T$ ,  $k = 1$  to  $K$  do
13:     $u_{t,k} \leftarrow (\nu_k + 1) / (\nu_k + ((O_t - \mu_k) / \sigma_k)^2)$ .
14:  end for
15:  CM-Step 1: update  $\mu_k, \sigma_k$  given  $\nu_k$ .
16:   $\pi_k \leftarrow \gamma_1(k)$  for all  $k$ .
17:  for  $k = 1$  to  $K$  do
18:     $w_{t,k} \leftarrow \gamma_t(k) u_{t,k}$ ;  $W_k \leftarrow \sum_t w_{t,k}$ ;  $G_k \leftarrow \sum_t \gamma_t(k)$ .
19:     $\mu_k \leftarrow \sum_t w_{t,k} O_t / W_k$ .
20:     $\sigma_k \leftarrow \max(\sqrt{\sum_t w_{t,k} (O_t - \mu_k)^2 / G_k}, 10^{-6})$ .
21:  end for
22:  CM-Step 2: update  $\nu_k$  by 1D golden-section search.
23:  for  $k = 1$  to  $K$  do
24:    Define  $Q_k(\nu) = \sum_t \gamma_t(k) \log t_{\nu}(O_t; \mu_k, \sigma_k)$ .
25:     $\nu_k \leftarrow \arg \max_{\nu \in [\nu_{\min}, \nu_{\max}]} Q_k(\nu)$  via 40-iteration golden-section search.
26:  end for
27:  Transition update.  $T_{ij} \leftarrow \sum_t \xi_t(i, j) / \sum_t \gamma_t(i)$ .
28:  Convergence check.  $\mathcal{L}^{(n)} \leftarrow \log \text{sumexp}_k \log \alpha_T(k)$ .
29:  if  $|\mathcal{L}^{(n)} - \mathcal{L}_{\text{prev}}| < \text{tol}$  then
30:    break
31:  end if
32:   $\mathcal{L}_{\text{prev}} \leftarrow \mathcal{L}^{(n)}$ .
33: end for
34: return  $\mathbf{T}, \{\mu_k, \sigma_k, \nu_k\}_{k=1}^K, \pi$ 

```

Algorithm 3 EM algorithm for continuous Laplace HMM (CHMM-L) parameter estimation

Require: Observations $O_{1:T}$, number of states K , tolerance tol , max_iter

Ensure: Transition matrix $\mathbf{T}$, emission parameters $\{\mu_k, b_k\}_{k=1}^K$, initial distribution π

```
1: Quantile-Based Initialization.
2: Sort observations  $O^{(\text{sort})} \leftarrow \text{sort}(O_{1:T})$  and partition into  $K$  equal chunks  $\{C_1, \dots, C_K\}$ .
3: for  $k = 1$  to  $K$  do
4:    $\mu_k \leftarrow \text{median}(C_k)$ ,  $b_k \leftarrow \max(\text{mean}(|C_k - \mu_k|), 10^{-6})$ .
5: end for
6:  $T_{ij} \leftarrow 1/K$ ;  $\pi_k \leftarrow 1/K$ ;  $\mathcal{L}_{\text{prev}} \leftarrow -\infty$ .
7: EM Loop.
8: for  $n = 1$  to  $\text{max\_iter}$  do
9:   E-Step: emission log-likelihoods  $\log B_{t,k} = -\log(2b_k) - |O_t - \mu_k|/b_k$ .
10:  Run the log-space forward-backward recursions (15)–(17) to obtain  $\gamma_t(k)$  and  $\xi_t(i, j)$ .
11:  M-Step (closed-form weighted Laplace MLE).
12:   $\pi_k \leftarrow \gamma_1(k)$  for all  $k$ .
13:  for  $k = 1$  to  $K$  do
14:     $W_k \leftarrow \sum_t \gamma_t(k)$ .
15:     $\mu_k \leftarrow \text{WeightedMedian}(O_{1:T}, \gamma_{1:T}(k))$  {cumulative-weight bracketing; linear interpolation at the  $W_k/2$  crossing.}
16:     $b_k \leftarrow \max(\sum_t \gamma_t(k) |O_t - \mu_k|/W_k, 10^{-6})$ .
17:  end for
18:  Transition update.  $T_{ij} \leftarrow \sum_t \xi_t(i, j) / \sum_t \gamma_t(i)$ .
19:  Convergence check.  $\mathcal{L}^{(n)} \leftarrow \log \text{sumexp}_k \log \alpha_T(k)$ .
20:  if  $|\mathcal{L}^{(n)} - \mathcal{L}_{\text{prev}}| < \text{tol}$  then
21:    break
22:  end if
23:   $\mathcal{L}_{\text{prev}} \leftarrow \mathcal{L}^{(n)}$ .
24: end for
25: return  $\mathbf{T}, \{\mu_k, b_k\}_{k=1}^K, \pi$ 
```

Algorithm 4 Viterbi decoding for the most likely hidden state sequence

Require: Observations $O_{1:T}$, Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\mu_k, \sigma_k\}, \pi)$

Ensure: Most probable state sequence $S_{1:T}^*$

```
1: for  $k = 1$  to  $K$  do
2:    $\log \delta_1(k) \leftarrow \log(1/K) + \log \mathcal{N}(O_1 \mid \mu_k, \sigma_k^2)$ .
3: end for
4: for  $t = 2$  to  $T$  do
5:   for  $j = 1$  to  $K$  do
6:      $\text{vals}(j) \leftarrow \log \delta_{t-1}(i) + \log T_{ij}$  for all  $i$ .
7:      $\log \delta_t(j) \leftarrow \max_i \text{vals}(i) + \log \mathcal{N}(O_t \mid \mu_j, \sigma_j^2)$ .
8:      $\psi_t(j) \leftarrow \arg \max_i \text{vals}(i)$ .
9:   end for
10: end for
11:  $S_T^* \leftarrow \arg \max_k \log \delta_T(k)$ .
12: for  $t = T - 1$  down to  $1$  do
13:    $S_t^* \leftarrow \psi_{t+1}(S_{t+1}^*)$ .
14: end for
15: return  $S_{1:T}^*$ 
```

Algorithm 5 CHMM simulation of synthetic excess growth rate paths

Require: Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\mu_k, \sigma_k\}, \boldsymbol{\pi})$, Number of steps M , Number of paths P

Ensure: Simulated return paths $\{\hat{G}_{1:M}^{(p)}\}_{p=1}^P$

```
1: for  $p = 1$  to  $P$  do
2:   Sample initial state  $S_1 \sim \text{Categorical}(\boldsymbol{\pi})$ .
3:   for  $t = 1$  to  $M$  do
4:     Emit observation  $\hat{G}_t^{(p)} \sim \mathcal{N}(\mu_{S_t}, \sigma_{S_t}^2)$ .
5:     if  $t < M$  then
6:       Transition to next state  $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t, \cdot})$ .
7:     end if
8:   end for
9: end for
10: return  $\{\hat{G}_{1:M}^{(p)}\}_{p=1}^P$ 
```

Algorithm 6 Cross-asset copula simulation via rank reordering

Require: Fitted per-asset CHMMs $\{\mathcal{M}_j\}_{j=1}^d$, correlation matrix Σ (PSD, unit diagonal), copula family $\mathcal{C} \in \{\text{Gaussian}, t_\nu\}$, path length T , number of paths P

Ensure: Cross-asset path tensor $\{\hat{G}_{j,t}^{(p)}\}$ of shape $T \times d \times P$

```
1: for  $p = 1$  to  $P$  do
2:   Draw copula sample  $\mathbf{U}^{(p)} \in [0, 1]^{T \times d}$  from  $\mathcal{C}(\Sigma)$ :
3:    $L \leftarrow \text{Cholesky}(\Sigma)$ ,  $Z \in \mathbb{R}^{T \times d} \sim \mathcal{N}(0, I_d)$  i.i.d.
4:    $Y \leftarrow ZL^\top$ 
5:   if  $\mathcal{C} = \text{Gaussian}$  then
6:      $\mathbf{U}^{(p)} \leftarrow \Phi(Y)$  componentwise
7:   else
8:     Draw  $W \sim \chi_\nu^2/\nu$  i.i.d. for  $t = 1, \dots, T$ ;  $X_{t,\cdot} \leftarrow Y_{t,\cdot}/\sqrt{W_t}$ 
9:      $\mathbf{U}^{(p)} \leftarrow t_\nu(X)$  componentwise
10:  end if
11:  for  $j = 1$  to  $d$  do
12:    Simulate  $\tilde{g}_{j,1:T}^{(p)}$  from  $\mathcal{M}_j$  via Algorithm 5.
13:    Compute order statistics  $\tilde{g}_{j,(1)}^{(p)} \leq \dots \leq \tilde{g}_{j,(T)}^{(p)}$ .
14:    Compute ranks  $r_{j,t}^{(p)} \leftarrow \text{rank}(U_{j,t}^{(p)})$  for  $t = 1, \dots, T$ .
15:     $\hat{G}_{j,t}^{(p)} \leftarrow \tilde{g}_{j,(r_{j,t}^{(p)})}^{(p)}$  for  $t = 1, \dots, T$ .
16:  end for
17: end for
18: return  $\{\hat{G}_{j,t}^{(p)}\}$ 
```

Algorithm 7 Single Index Model (SIM) construction and simulation

Require: In-sample return matrix $\mathbf{R} \in \mathbb{R}^{T \times d}$, market-column index m , market CHMM $\mathcal{M}_m$, path length T , number of paths P

Ensure: Cross-asset path tensor $\{\hat{G}_{j,t}^{(p)}\}$ of shape $T \times d \times P$

- 1: **OLS fit.** For $j \neq m$:
 - 2: $\hat{\beta}_j \leftarrow \text{Cov}(\mathbf{R}_{:,j}, \mathbf{R}_{:,m}) / \text{Var}(\mathbf{R}_{:,m})$
 - 3: $\hat{\alpha}_j \leftarrow \overline{\mathbf{R}_{:,j}} - \hat{\beta}_j \overline{\mathbf{R}_{:,m}}$
 - 4: $\hat{\eta}_{j,t} \leftarrow R_{t,j} - \hat{\alpha}_j - \hat{\beta}_j R_{t,m}$, $R_j^2 \leftarrow 1 - \sum_t \hat{\eta}_{j,t}^2 / \sum_t (R_{t,j} - \overline{\mathbf{R}_{:,j}})^2$
 - 5: **for** $p = 1$ to P **do**
 - 6: Simulate market path $G_{m,1:T}^{(p)}$ from $\mathcal{M}_m$ via Algorithm 5.
 - 7: $\hat{G}_{m,t}^{(p)} \leftarrow G_{m,t}^{(p)}$.
 - 8: **for** $j \neq m$ **do**
 - 9: Draw residual indices $\{\iota_t\}_{t=1}^T \sim \text{Uniform}(\{1, \dots, T\})$.
 - 10: $\hat{G}_{j,t}^{(p)} \leftarrow \hat{\alpha}_j + \hat{\beta}_j G_{m,t}^{(p)} + \hat{\eta}_{j,\iota_t}$.
 - 11: **end for**
 - 12: **end for**
 - 13: **return** $\{\hat{G}_{j,t}^{(p)}\}$
-